Polynomial Semantics of Dependent Type Theory

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Abstract

We propose a new algebraic semantics for dependently typed algebraic theories inspired by the theory of polynomial functors. We first introduce a notion of *dependent polynomial*, where operation symbols can have non-trivial dependencies in their arities. Dependent polynomials can be viewed spans in a certain presheaf category and inherit a natural notion of composition. Monoids for this composition are called *dependent polynomial monads*, and represent our axiomatization of a collection of dependently typed terms closed under substitution.

An advantage of this perspective is that our dependent polynomial monads come with a nautral notion of bimodule and the category of dependent polynomials and their bimodules enjoys many nice structural features which follow from general theory. Our dependent polynomial monads lack, however, a means of substituting terms in types. We introduce the notion of a *substitution structure* to account for this difference. A dependent polynomial monad equipped with a well-behaved substitution structure is called a *polynomial theory*, and we show that polynomial theories are equivalent to the B-systems of Voevodsky.

1 Introduction

We propose a new semantic framework for dependently typed algebraic theories. There are, of course, many such models already present in the literature: categories with attributes [5], categories with families [7], comprehension categories [11], natural models [3], and B-systems [16] form a representative sample. (See [2] for nice overview of the various comparisions and equivalences which are known among them). Nonetheless, we feel that the model we propose here provides an original point of view on the subject and in doing so, makes contact with and takes its inspiration from, two other well established notions in categorical logic. Specifically,

(1) The theory of *polynomial functors*
(2) The theory of *generalized multicategories*

Let us take comment on each of these in turn.

Polynomial Functors Recall that an indexed polynomial functor is determined by a diagram of sets of the form

$$I \xleftarrow{\tau} E \xrightarrow{\rho} B \xrightarrow{\gamma} J$$

In the most common case, $I = J$ and we will assume this in the discussion to follow (though we will have more to say about the inhomogeneous case later). We think of such a diagram as a typed algebraic signature: I is our set of *type symbols* and B our set of *operator symbols*. Each operator $b \in B$ is assigned a return type by the map γ and the fiber E_b of the map ρ is interpreted as the *arity* of the operator. Finally, each of its inputs is assigned a type by the map τ. Indeed, we can even picture such an element in a type theoretic form as:

$$(x_0 : i_0) \dots (x_n : i_n) \vdash b(x_0, ..., x_n) : i$$

Here we may think of the variables $x_i \in E_b$ as enumerating the elements of the fiber E_b, their types are assigned by τ (that is, $\tau(x_k) = i_k$) and the type of the operator b is given by γ (that is, $\gamma(b) = i$). In other words, we can think of the polynomial as a collection of generating judgements for a simply typed algebraic theory. In order to complete the definition of our theory, we should explain how these operations interact with each other, i.e. the relations they satisfy. When our signature is described by a polynomial in this way, such a description can be given by endowing the polynomial with a monad structure, and such a monad is known as a *polynomial monad*. A number of commonly occuring mathematical structures can be understood as algebras for a polynomial monad: monoids, planar operads, etc.

Inspecting this notion of polynomial, however, we see that the collection of types is modelled by a mere set, namely I. In particular, there does not arise the opportunity to have dependently typed operations as the set I does not record any information about how types might depend on each other. A point of departure for the current work is the question of whether the standard notion of polynomial can be generalized in some way so as to accommodate *dependently typed signatures* and such that we can understand dependently typed algebraic theories as these more general polynomials equipped with a relevant type of monadic structure. The answer turns out to be yes, and this description brings us directly into contact with the other theme of this work.

Generalized Multicategories Multicategories were introduced by Joachim Lambek in [12] as part of his attempt to give a categorical account of deductive systems. In a multicatgory, morphisms are allowed to have a *list* of objects in their domain, and the central idea is to view a morphism

$$\phi : [A, B, C] \to D$$

as a witness of the entailment relation

$$A, B, C \vdash D$$

of a deductive system, with composition in the multicategory implementing a form of the cut rule. Our second, closely related goal, then, is to give an account of the algebraic semantics for dependent types which arises directly from this same multi-categorical intuition, with the main issue being how to account for type dependency.

In order to do so, we will need to consider a generalized notion of multicategory. Luckily, the theory of multicategories can be carried out in a quite general setting, and our approach falls nicely within this framework. We will in particular make use of the presentation in [13] which gives an account of a general theory of multicategories starting from a cartesian category $\mathscr{C}$ equipped with a cartesian monad T and defines the notion of a T-multicategory as a monoid

in T-spans. In the case at hand, T will be the monad of *contexts* in a category of *type structures* meant to model type dependency. We will refer to T-spans as *dependent polynomials* and the associated T-monoids as *dependent polynomial monads*.

Our dependent polynomial monads and their bimodules can be organized into a special kind of double category, an *equipment* or *framed bicategory* [15], and we believe this is an interesting object in its own right. Studying the analogous structure in the case of operads and their bimodules was topic of [9]. Furthermore, multiple authors ([6], [8]) have shown that this construction can be done quite generally, and we make use of their results here.

That such a well-behaved and natural theory of bimodules exists for our dependent polynomial monads was another motivation for our approach. We think this concept deserves to be more generally known in type theory circles. If dependent type theory is thought of as an algebra of judgements, then a dependent bimodule can be imagined as a collection of judgements:

$$(x_0 : A_0) \ldots (x_n : A_n\, x_0 \ \ldots\ x_{n-1}) \vdash b(x_0, \ldots, x_n) : B$$

where the terms b have *input types* in one theory, but *output types* in another, with these terms closed under substitution from the domain theory on the left, and substitution in the codomain theory on the right. Any homomorphism of theories gives rise to such a bimodule, but many functors between models of dependently typed theories arise from bimodules, but not homomorphisms.

Dependent polynomial monads turn out to be somewhat simpler objects than what we find in usual algebraic semantics of dependent types. The reason is that the structure we obtain captures the notion of a collection of dependently typed terms closed under a composition operation, but the types and their dependencies are fixed in advance. In particular, the resulting structure has no notion of substituting a term *into* a type. This is partially by design: part of the novelty of this work is exactly to show that we can factor our semantics in this way.

On the other hand, realistic dependetly typed algebraic theories of course *do* allow for terms to be substituted in types. We recover this concept by introducing the notion of a *substitution structure* on a dependent polynomial monad, and we call the resulting notion a *polynomial theory*. Our main result is now that the category of polynomial theories is equivalent to the category of B-systems of Voevodsky, thus demonstrating the compatibility of our approach with those already in the literature.

Finally, a note about our style in this article. We begin, in Section 2 with an exposition of the general setup in categorical style. This lets us leverage the existing work on the subject and explain how our model fits into a general pattern and connects to the topics outlined in this introduction. After setting up the abstract framework, however, we will then unfold the relevant definitions in a style much closer to dependent type theory itself. There are several reasons for this shift in style. A first is simply that we have partially formalized the results here, and the presentation draws on this formalization. Second, our main result, an equivalence with the theory of B-systems, requires us to pick a presentation of these structures to work with, and we much prefer the coninductive description of these objects suggested in [2] and formalized in [14]. Our dependent polynomials *also* admit a very natural coinductive presentation, and we found this to be the most expedient way to describe the equivalence. Said otherwise, part of the contribution of this work is translating this particular case of the general theory of T-multicategories into type theory, and developing the consequences from that point of view.

2 Dependent Polynomial Monads and their Bimodules

In this section, we give a categorical description of our dependent polynomials and their monads. The presentation will be somewhat terse, relying heavily on the literature, since we will spell out all the definitions explicitly in the sections to follow.

To begin with, we will work in the category of *type structures*. These are presheaves on the natural numbers viewed as a category with the usual $\leq$ partial order structure on $\mathbb{N}$. That is, we defined

$$\mathtt{TyStr} := \mathsf{Set}^{\mathbb{N}^{op}_{\leq}}$$

Concretely, then, a type structure $\mathbb{T}$ is a diagram of sets

$$\cdots \longrightarrow \mathbb{T}_n \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_1} \mathbb{T}_1 \xrightarrow{\partial_0} \mathbb{T}_0$$

where we write $\mathbb{T}_i = \mathbb{T}(i)$ and we think of the elements of $\mathbb{T}_i$ as types dependent on i variables. Iterating the boundary maps ∂_i then tells us all the types on which a given type depends. As the category $\mathtt{TyStr}$ is a presheaf topos, it enjoys many nice categorical properties which will be useful in what follows: it is complete and cocomplete (in particular, cartesian) and colimts are stable by pullback.

There is a monad $Ctx : \mathtt{TyStr} \to \mathtt{TyStr}$ sending a fixed type structure $\mathbb{T}$ to its collection of *dependent contexts*, where a context in a type structure $\mathbb{T}$ is simply a finite list of types $A_0, \ldots, A_n$ such each $A_i \in \mathbb{T}_i$ and $\partial_{n-1} A_n = A_{n-1}$. To get an intuition for this functor, suppose that we have an some collection of types $A_i \in \mathbb{T}_i$ in a type structure $\mathbb{T}$ with with $\partial_{i-1} A_i = A_{i-1}$. We'll denote such a configuration as

$$\cdots \Rightarrow A_4 \Rightarrow A_3 \Rightarrow A_2 \Rightarrow A_1 \Rightarrow A_0$$

(The reader may wish to think of this as an actual diagram in the category of elements of $\mathbb{T}$). Now, given such elements of $\mathbb{T}$, a typical element of $Ctx(\mathbb{T})$ can be visualized as

$$\begin{array}{cc}
\mathbb{T} & Ctx(\mathbb{T}) \\
\vdots & \vdots \\
\Downarrow & \Downarrow \\
A_3 & \Gamma_3 = [A_4, A_5] \\
\Downarrow & \Downarrow \\
A_2 & \Gamma_2 = [\,] \\
\Downarrow & \Downarrow \\
A_1 & \Gamma_1 = [A_3] \\
\Downarrow & \Downarrow \\
A_0 & \Gamma_0 = [A_0, A_1, A_2]
\end{array}$$

In summary, contexts must be extended in a way compatible with the dependency structure of $\mathbb{T}$, and a context dependent on another is one which starts with a type dependent on the final type of the context it lives over (if it is non-empty). But note that types "higher" in the dependency stucture $\mathbb{T}$ can appear lower in $Ctx(\mathbb{T})$ as they are gathered together into contexts.

Formally, we define

$$\begin{aligned} Ctx(\mathbb{S})(n) = \{([A_0, \ldots, A_{i_0}], \\ [A_{i_0+1}, \ldots, A_{i_1}], \ldots, \\ [A_{i_{n-1}+1}, \ldots, A_{i_n}]) \\ \mid A_i \in \mathbb{T}_i, \partial_{i-1} A_i = A_{i-1}\} \end{aligned}$$

with the projection map ∂_i acting on these tuples of lists by forgetting the last element of the tuple.

Definition 2.1. A *dependent polynomial* $P : \mathbb{S} \nrightarrow \mathbb{T}$ is a span of the form

$$\begin{tikzcd} & P \arrow[dl] \arrow[dr] & \\ Ctx(\mathbb{S}) & & \mathbb{T} \end{tikzcd}$$

in the category `TyStr`.

Unpacking this definition, one finds that a dependent polynomial consists of a collection of "terms" whose domains are *contexts* in the domain type structure, and whose codomains *types* in the codomain structure. These terms themselves carry information about the terms on which they depend in order to be well typed. Schematically, we can picture, for example, and element of $P(2)$ as:

$$\begin{tikzcd} {[A_2]} \arrow[r, "p_2"] \arrow[d, Rightarrow] & B_2 \arrow[d, Rightarrow] \\ {[]} \arrow[r, "p_1"] \arrow[d, Rightarrow] & B_1 \arrow[d, Rightarrow] \\ {[A_0, A_1]} \arrow[r, "p_0"] & B_0 \end{tikzcd}$$

Corresponding to the judgements

$$\begin{aligned} (a_0 : A_0)(a_1 : A_1\, a_0) &\vdash p_0\, a_0\, a_1 : B_0 \\ (a_0 : A_0)(a_1 : A_1\, a_0) &\vdash p_1\, a_0\, a_1 : B_1\, (p_0\, a_0\, a_1) \\ (a_0 : A_0)(a_1 : A_1\, a_0)(a_2 : A_2\, a_0\, a_1) & \\ &\vdash p_2\, a_0\, a_1\, a_2 : B_2\, (p_0\, a_0\, a_1)\, (p_1\, a_0\, a_1) \end{aligned}$$

We view these types of spans as the dependent analog of ordinary polynomials, as described in the introduction, that is, a collection of generating, dependently typed term judgements. In addition to the types carrying dependency information, we see that in such a polynomial, we also need *terms* to carry dependency information since their types may not be valid before the introduction of terms they depend on.

Our definition corresponds to what is called a *T-span* in [13] [6]. In particular we obtain a double category `DepPoly`, (in fact, an equipment or framed bicategory) of such spans pictured as

$$\begin{tikzcd} \mathbb{S} \arrow[r, "P", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow"] & \mathbb{T} \arrow[d, "g"] \\ \mathbb{U} \arrow[r, "Q"', "\shortmid"{marking}] & \mathbb{V} \end{tikzcd}$$

in which the horizontal arrows are dependent polynomials and the vertical arrows are maps in the category `TyStr` of type structures, and the 2-cells are maps of spans. See [13, Definition 4.2.1] or [6, Definition 4.1] for more details on this construction.

Definition 2.2. A *dependent polynomial monad* is a monoid for the horizontal composition in the double category `DepPoly`.

General theory tells us that, in good circumstances, once we have a double category as we have just constructred, the monoids internal to that double category, together with their homomorphisms and bimodules *again* form a double category. A precise statement is [8, Proposition 2.3.3]:

Proposition 2.3. *Let C be a double category with stable local reflexive coequalisers. There is a double category* $\mathtt{Bim}(C)$ *with monoids as objects, monoid morphisms as vertical maps, monoid bimodules as horizontal maps and bimodule 2-morphisms as 2-morphisms. Furthermore,* $\mathtt{Bim}(C)$ *itself has stable local reflexive coequalisers.*

In order to apply this construction, we simply have to verify that `DepPoly` indeed has stable, local reflexive coequalizers. This follows from the fact that the Ctx functor preserves reflexive coequalizes as we now show. A first observation is:

Lemma 2.4. *The Ctx monad is familially representable ([4])*

Proof. According to [10, Proposition 44], it is sufficient to exhibit a type structure $\mathbb{B}$ and a functor $\mathbb{E} : el(\mathbb{B}) \to \mathtt{TyStr}$ such that

$$Ctx(\mathbb{S})(n) \simeq \coprod_{b \in \mathbb{B}(n)} Hom(\mathbb{E}(n, b), \mathbb{S})$$

Here, $el(\mathbb{B})$ denotes the category of elements.

The relevant $\mathbb{B}$ can be desribed as

$$\mathbb{B}(n) = \{(l_0, \ldots, l_n) \mid l_i \in \mathbb{N}\}$$

that is, where $\mathbb{B}(n)$ consists of $n + 1$ tuples of natural number. Moreover $\mathbb{E}$ can be given as

$$\mathbb{E}(n, (l_0, \ldots, l_n))(k) = \begin{cases} \star & \text{if } k < l_0 + \cdots + l_n \\ \emptyset & \text{otherwise} \end{cases}$$

The reader can easily check that this choice of $\mathbb{B}$ and $\mathbb{E}$ produces the definition of Ctx given above. □

Corollary 2.5. *Ctx preserves reflexive coequalizers.*

Proof. From the explicit description, we see that the value of the functor on any $n \in \mathbb{N}$ is given as a coproduct of representables whose domain one of the *finite* presheafs $\mathbb{E}(n, b)$. These functors preserved sifted colimits (see, for example, [1]) of which reflexive coequalizers are prototypical example. □

We record, then, the output of these constructions as the main theorem of this section:

Theorem 2.6. *There is a framed bicategory* `DepPolyMnd` *with 0-cells the dependent polynomial monads, vertical 1-cells homomorphisms of such monads, horizontal 1-cells bimodules and 2-cells the bimodule homomorphisms.*

Proof. We apply Proposition 2.3. To say that our double category has stable reflexive coequalizers is to say that, for fixed type structures $\mathbb{S}$ and $\mathbb{T}$, the horizontal category $\mathtt{DepPoly}(\mathbb{S}, \mathbb{T})$ has reflexive coequalizers which are preserved by horizontal composition in each variable. Now, the category $\mathtt{DepPoly}(\mathbb{S}, \mathbb{T})$ is in fact a slice category of a presheaf topos, namely

$$\mathrm{Set}^{\mathbb{N}_{\leq}^{op}} / Ctx(\mathbb{S}) \times \mathbb{T}$$

and therefore again a presheaf topos. In particular, it has reflexive coequalizers. To see that they are preserved by composition, we recall that the composition of spans $P : \mathbb{S} \nrightarrow \mathbb{T}$ and $Q : \mathbb{T} \nrightarrow \mathbb{U}$ is given as (see [13, Definition 4.2.1]) by the diagram

$$\begin{tikzcd}
 & & & P \circledcirc Q \arrow[dl] \arrow[dr] & & \\
 & & Ctx(P) \arrow[dl] \arrow[dr] & & Q \arrow[dl] \arrow[dr] & \\
 & Ctx(Ctx(\mathbb{S})) \arrow[dl] & & Ctx(\mathbb{T}) & & \mathbb{U} \\
Ctx(\mathbb{S}) & & & & &
\end{tikzcd}$$

And we see clearly from the diagram that a reflexive colimit in the P variable will be preserved by Corollary 2.5 combined with the fact that in a presheaf category, colimits are stable by pullback and similarly for Q. □

3 Dependent Polynomials Coinductively

In this section, we revisit the definitions of the previous section in a coinductive style, spelling out explicitly the defining data of each of the objects in question.

3.1 Type Structures

We start with type structures. As we have seen, these are essentially just towers of sets, containing types which can depend on the types below them in the tower. In the language of dependent type theory, however, we have a natural *indexed* presentation, in contrast with the *fibered* one of the previous section.

Definition 3.1. A *type structure* $\mathbb{T}$ consists of a set $Ty_{\mathbb{T}}$ of types together with, for each $A \in Ty_{\mathbb{T}}$ a type structure $\mathbb{T}//A$, of types dependent on A.

Type structures have a natural notion of morphism, which can also be defined coinductively:

Definition 3.2. Let $\mathbb{S}$ and $\mathbb{T}$ be type structures, a morphism $f : \mathbb{S} \to \mathbb{T}$ is defined coinductively as follows:

(1) a morphism $f_{Ty} : Ty_{\mathbb{S}} \to Ty_{\mathbb{T}}$
(2) for each $A \in Ty_{\mathbb{S}}$, a morphism of type structures $f//A : \mathbb{S}//A \to \mathbb{T}//f_{Ty}(A)$

Our next step is to build contexts in this setting. Analogous to the usual definition, these are defined inductively.

Definition 3.3. Let $\mathbb{T}$ be a type structure, then the set of *contexts in* $\mathbb{T}$ is inductively generate by the following two clauses

(1) there is an empty context $\epsilon \in Ctx_{\mathbb{T}}$
(2) for each $A : Ty(\mathbb{T})$ and $\Gamma \in Ctx_{\mathbb{T}//A}$ we have an extended context $A \triangleright \Gamma \in Ctx_{\mathbb{T}}$

Remark 3.1. For each type A in a type structure $\mathbb{T}$, we have a context consisting of just that type, built as $A \triangleright \epsilon$. When it will not cause any confusion, we will abuse notation and write A for both the type and the context consisting of just that type.

Construction 3.1. Let $\mathbb{T}$ be a type structure, then each context $\Gamma \in Ctx_{\mathbb{T}}$ induces a type structure $\mathbb{T}//\Gamma$ as follows:

- if $\Gamma = \epsilon$ then $\mathbb{T}//\Gamma = \mathbb{T}$
- otherwise, for $\Gamma = A \triangleright \Delta$ we set $\mathbb{T}//\Gamma = (\mathbb{T}//A)//\Delta$.

Remark 3.2. Since the operation $\mathbb{T}//\Gamma$ is nothing more that iterating $\mathbb{T}//_$ over the types in Γ, we abuse notation and use $_//_$ for both types and contexts.

Definition 3.4. For a type structure $\mathbb{T}$, the type structure of *contexts of* $\mathbb{T}$, denoted $Ctx(\mathbb{T})$ is defined by

$$Ty(Ctx(\mathbb{T})) = Ctx_{\mathbb{T}}$$
$$Ctx(\mathbb{T})//\Gamma = Ctx(\mathbb{T}//\Gamma)$$

Construction 3.2. Let $\mathbb{T}$ be a type structure, let $\Gamma \in Ctx_{\mathbb{T}}$ and $\Delta \in Ctx_{\mathbb{T}//\Gamma}$, then there exists a context $\Gamma +\!\!+ \Delta \in Ctx_{\mathbb{T}}$. We do this by recursion on Γ: If $\Gamma = \epsilon$ then $\Gamma +\!\!+ \Delta = \Delta$. If $\Gamma = A \triangleright \Gamma'$ then $\Gamma +\!\!+ \Delta = A \triangleright (\Gamma' +\!\!+ \Delta)$.

Remark 3.3. The monad structure for Ctx is essentially that of the *List* monad on Set, but levelwise: μ can be obtained from the operation $+\!\!+$, while η is given by the inclusion of singletons.

3.2 Dependent Polynomials

From here we move on to the definition of dependent polynomials.

Definition 3.5. Let $\mathbb{S}$ and $\mathbb{T}$ be type structures, a *dependent polynomial* $P : \mathbb{S} \nrightarrow \mathbb{T}$ is defined coinductively to consist of

(1) a set $P_{\Gamma,A}$ of terms, for each context $\Gamma \in Ctx_{\mathbb{S}}$ and type $A \in Ty(\mathbb{T})$
(2) for each $a \in P_{\Gamma,A}$ a dependent polynomial $P \uparrow a : \mathbb{S}//\Gamma \nrightarrow \mathbb{T}//A$

Our next goal is to define the composition of two dependent polynomials. To do this, we first need to define the notion of substitutions inside a polynomial. This will be done inductively over the structure of contexts akin to the usual definition of substitutions.

Definition 3.6. Let $\mathbb{S}$ and $\mathbb{T}$ be type structures and $P : \mathbb{S} \nrightarrow \mathbb{T}$ a dependent polynomial. Given contexts $\Gamma \in Ctx_{\mathbb{S}}$ and $\Delta \in Ctx_{\mathbb{T}}$, we define the set of substitutions $\Gamma \mapsto \Delta$ inductively as follows:

(1) For any $\Gamma \in Ctx_{\mathbb{S}}$ we have $\bullet\Gamma \in \Gamma \mapsto \epsilon$
(2) For any $A \in Ty_{\mathbb{T}}$, $a \in P_{\Gamma,A}$, $\Gamma' \in Ctx_{\mathbb{S}//\Gamma}$, $\Delta \in Ctx_{\mathbb{T}//A}$ and σ a substitution in $P \uparrow a$ from Γ' to Δ, we have a substitution $\langle a, \sigma \rangle \in (\Gamma +\!\!+ \Gamma') \mapsto (A \triangleright \Delta)$

Construction 3.3. Let $\mathbb{S}$ and $\mathbb{T}$ be type structures and let $P : \mathbb{S} \nrightarrow \mathbb{T}$ be a dependent polynomial. Further let Γ be a context in $\mathbb{S}$, Δ be a context in $\mathbb{T}$ and $\sigma \in \Gamma \mapsto \Delta$ be a substitution in P. Then σ induces a dependent polynomial $P \uparrow \sigma$ on $\mathbb{S}//\Gamma$ and $\mathbb{T}//\Delta$. This be done by recursion on σ:

- If $\sigma = \bullet\Gamma$ then we set $(P \uparrow \sigma) \uparrow a = P \uparrow a$.
- If $\sigma = \langle t, \sigma' \rangle$ then we choose $(P \uparrow \sigma) = (P \uparrow t) \uparrow \sigma'$.

The notation here is mildly abusive, as in Remark 3.2, but should not cause confusion.

Construction 3.4. The concatenation operation $+\!\!+$ on contexts extends to substitutions in a nautral way: let $\mathbb{S}$ and $\mathbb{T}$ be type structures and let $P : \mathbb{S} \nrightarrow \mathbb{T}$ be a dependent polynomial, Γ a context in $\mathbb{S}$, Δ be a context in $\mathbb{T}$ and $\sigma \in \Gamma \mapsto \Delta$ be a substitution in P. Additionally, let Γ' be a context in $\mathbb{S}//\Gamma$, Δ' be a context in $\mathbb{T}//\Delta$ and let $\sigma' \in \Gamma' \mapsto \Delta'$ be a substitution in $P \uparrow \sigma$. Then we can define a substitution $\sigma +\!\!+ \sigma' \in (\Gamma +\!\!+ \Gamma') \mapsto (\Delta +\!\!+ \Delta')$ in P by recursion on σ:

- When $\sigma = \bullet\Gamma$ we set $\bullet\Gamma +\!\!+ \sigma' = \sigma'$
- And when $\sigma = \langle t, \sigma'' \rangle$ we set $\langle t, \sigma'' \rangle +\!\!+ \sigma' = \langle t, \sigma'' +\!\!+ \sigma' \rangle$

Remark 3.4. Let $\mathbb{S}$, $\mathbb{T}$ and P be as above, then for each context Γ in $\mathbb{S}$ and type A in $\mathbb{T}$, every term a in $P_{\Gamma,A}$ induces a substitution $\Gamma \mapsto A$ in P defined as $\langle a, \bullet\epsilon \rangle$. When the meaning is clear from the context, we will often conflate a term and the substitution it corresponds to.

With substitutions in hand, we can now define composition of polynomials. The reader may wish to compare with the categorical description given in the proof of Theorem 2.6

Definition 3.7. Let $\mathbb{S}, \mathbb{T}$ and $\mathbb{R}$ be type structures and let $P : \mathbb{S} \nrightarrow \mathbb{T}$ and $Q : \mathbb{T} \nrightarrow \mathbb{R}$ be dependent polynomials. Then we can define $P \circledcirc Q$ coinductively as follows:

(1) Let Γ be a context in $\mathbb{S}$ and A a type in $\mathbb{R}$, then $(P \circledcirc Q)_{\Gamma,A}$ is the set of tuples (Δ, σ, t) where Δ is a context in $\mathbb{T}$, $\sigma \in \Gamma \mapsto \Delta$ is a substitution in P and t is a term in Δ and A in Q, so $t \in Q_{\Delta,A}$.
(2) Let $a = (\Delta, \sigma, t)$, be a type in $P \circledcirc Q$, then $(P \circledcirc Q) \uparrow a = (P \uparrow \sigma) \circledcirc Q \uparrow t$

Remark 3.5. As is reflected in the notation above, polynomial composition will be written in diagrammatic order, which explains our choice of $\circledcirc$ to denote this composition.

It will be useful to continue using a diagrammatic notation for working with terms and substitutions in polynomials, building on the description given in Section 2. We will denote both terms and substitutions as horizontal arrows, and continue to denote dependency with the double arrow $\Downarrow$. Though in what follows we will often be working in a single polynomial P, this diagrammatic notation works to depict terms and substitutions in the composite $P \circledcirc Q$ as well. For example, the the diagram

$$\begin{tikzcd}
\Gamma \arrow[r, "a"] \arrow[d, Rightarrow] & A \arrow[r, "b"] \arrow[d, Rightarrow] & B \arrow[d, Rightarrow] \\
\Delta \arrow[r, "\sigma", "P"'] & \Lambda \arrow[r, "c", "Q"'] & C
\end{tikzcd}$$

displays two terms of the composite polynomial $P \circledcirc Q$. A term $a \in \Gamma \mapsto A$ over a substitution $\sigma \in \Delta \mapsto \Lambda$ in P as well as a term $b \in A \mapsto B$ over a term $c \in \Lambda \mapsto C$ in Q. The intermediate contexts Λ and A are exactly the data needed to make this a well defined element of $P \circledcirc Q$.

Remark 3.6. While they are easily pictured in diagrams, including the context explicitly in tuples denoting elements of $P \circledcirc Q$ can often be notationally heavy. And as the contexts are often clear from the substitution, we will often omit the context Δ, writing simply (σ, t) for a term of $P \circledcirc Q$.

As described in Section 2, dependent polynomials are naturally organized into a double category with the 2-cells morphisms of polynomials which our task is now to define.

$$\begin{tikzcd}
\mathbb{S} \arrow[r, "P", "\shortmid"{marking}] \arrow[d, "f"'] \arrow[dr, phantom, "\Downarrow"] & \mathbb{T} \arrow[d, "g"] \\
\mathbb{S}' \arrow[r, "Q"', "\shortmid"{marking}] & \mathbb{T}'
\end{tikzcd}$$

To do so, fix $f : \mathbb{S} \to \mathbb{S}'$ and $g : \mathbb{T} \to \mathbb{T}'$ as in the diagram above.

Definition 3.8. A morphism of dependent polynomials relative to f and g is defined coinductively as

(1) for each context $\Gamma \in Ctx_{\mathbb{S}}$ and type $A \in Ty(\mathbb{T})$, a map $k_{\Gamma,A} : P_{\Gamma,A} \to Q_{f(\Gamma),g(A)}$
(2) for each context $\Gamma \in Ctx_{\mathbb{S}}$ and type $A \in Ty(\mathbb{T})$ and term $a \in P_{\Gamma,A}$, a morphism of dependent polynomials $k \uparrow a : P \uparrow a \to Q \uparrow k_{\Gamma,A}(a)$ relative to $f//\Gamma$ and $g//A$.

In many cases (such as the definition of monad below), f and g will be the identity, and we omit them from the notation.

Finally, as it will play a role in what follows, we record here the definition of the polynomial which serves as the identity for horizontal composition:

Definition 3.9. Let $\mathbb{T}$ be a type structure then we define the identity dependent polynomial $\mathbb{1}_{\mathbb{T}} : \mathbb{T} \nrightarrow \mathbb{T}$ coinductively in the following way:

(1) For each type A in $\mathbb{T}$ the set of terms $\mathbb{1}_{\mathbb{T}_{A,A}}$ is just the singleton set $\{\mathbb{1}_A\}$.
(2) For each context $\Gamma \neq A$ the set $\mathbb{1}_{\mathbb{T}_{\Gamma,A}}$ is empty.
(3) $\mathbb{1}_{\mathbb{T}} \uparrow \mathbb{1}_A = \mathbb{1}_{\mathbb{T}//A}$.

We will write $\mathbb{1}_A$ and $\mathbb{1}_\Gamma$ for the identity term and identity substitution whose existence are guaranteed by the definition, the latter being defined by induction on Γ.

3.3 Dependent Polynomial Monads

Definition 3.10. Let $\mathbb{T}$ be a type structure, a *polynomial monad* M on $\mathbb{T}$ consists of a dependent polynomial $P : \mathbb{T} \nrightarrow \mathbb{T}$ together with two morphisms of dependent polynomials: A unit $\eta : \mathbb{1}_{\mathbb{T}} \to P$ and a multiplication $\mu : P \circledcirc P \to P$ where it holds that:

- $\mu(\mu(P \circledcirc P) \circledcirc P) = \mu(P \circledcirc \mu(P \circledcirc P))$
- $\mu(\eta(\mathbb{1}_{\mathbb{T}}) \circledcirc P) = P = \mu(P \circledcirc \eta(\mathbb{1}_{\mathbb{T}}))$

A monad structure provides us with a means of postcomposing a substitution with a term. Iterating gives us a general notion of the composition of substitutions in a dependent polynomial monad:

Remark 3.7. The unit map $\eta : \mathbb{1}_{\mathbb{T}} \to P$ provides us with, for every type $A \in \mathbb{T}$ with a term $\eta_A : A \to A$ in the polynomial P by setting

$$\eta_A = k_{A,A}(\mathbb{1}_A)$$

Similarly, by induction, for any context $\Gamma \in Ctx(\mathbb{T})$, we have a substitution $\eta_\Gamma : \Gamma \mapsto \Gamma$ in P.

Lemma 3.11. *Let $\mathbb{T}$ be a type structure and let $M = (P, \mu, \eta)$ be a polynomial monad on $\mathbb{T}$. Further let Γ, Δ and Ψ be contexts in $\mathbb{T}$, let $\sigma \in \Gamma \mapsto \Delta$ and $\tau \in \Delta \mapsto \Psi$ be substitutions in P, then we can define a composition of substitutions $\sigma \circledcirc \tau$.*

Proof. We do this by recursion on τ: If $\tau = \bullet\Delta$ then $\sigma \circledcirc \tau = \bullet\Gamma$. If $\tau = \langle t, \tau' \rangle$, where we have $t \in P_{\Delta',T}$ together with a substitution $\tau' \in \Delta'' \mapsto \Psi'$, where $\Delta = \Delta' +\!\!+ \Delta''$ and $\Psi = T \triangleright \Psi'$. Further since substitutions are inductively defined as lists of terms, there exists an initial segment $\sigma' \in \Gamma' \mapsto \Delta'$ of σ together with a substitution $\sigma'' \in \Gamma'' \mapsto \Delta''$ where $\Gamma = \Gamma' +\!\!+ \Gamma''$ and $\sigma = \sigma' +\!\!+ \sigma''$. Then we take $\sigma \circledcirc \tau = \langle \mu(\sigma', t), \sigma'' \circledcirc \tau' \rangle$. □

Remark 3.8. We again use $\circledcirc$ to denote the composition of substitutions, since we also compose them in the diagrammatic order and therefore want to distinguish this composition from the usual composition $\circ$.

Remark 3.9. From now on we will apply μ directly to substitutions, since we can naturally extend their application to substitutions using Lemma 3.11 and the previous remark.

We now show that polynomial monads can be lifted along the slice operation on type structures. This can be achieved by lifting the monad operations along the identity terms and exploiting the monad laws which show that identity terms composed among themselves.

Proposition 3.12. *Let $\mathbb{T}$ be a type structure and let $M = (P, \mu, \eta)$ be a polynomial monad on $\mathbb{T}$. Further let A be a type in $\mathbb{T}$, then there exists a monad $M//A$ on $\mathbb{T}//A$.*

Proof. We choose $P//A = P \uparrow \eta_A$, $\eta//A = \eta \uparrow \eta_A$ and $\mu//A = \mu \uparrow (\eta_A, \eta_A)$. The fact that the monad laws are bisimulations, together with the unit law implies that we can reconstruct them for $P \uparrow \eta_A$. We show this only for the unit law, the associative law follows analogously:

$$\begin{aligned}
&\mu//A(\eta//A(\mathbb{1}_{\mathbb{T}} \uparrow \eta_A) \circledcirc P \uparrow \eta_A)\\
=&\mu \uparrow (\eta_A, \eta_A)(\eta \uparrow \mathbb{1}_A(\mathbb{1}_{\mathbb{T}} \uparrow \mathbb{1}_A) \circledcirc P \uparrow \eta_A)\\
=&\mu \uparrow (\eta_A, \eta_A)(\eta \uparrow \mathbb{1}_A(\mathbb{1}_{\mathbb{T}//A}) \circledcirc P \uparrow \eta_A)\\
=&\mu \uparrow (\eta_A, \eta_A)(\eta(\mathbb{1}_{\mathbb{T}}) \circledcirc P) \uparrow (\eta_A, \eta_A)\\
=&\mu(\eta(\mathbb{1}_{\mathbb{T}}) \circledcirc P) \uparrow \eta_A
\end{aligned}$$

Where the same argument applies to $\mu(P \circledcirc \eta(\mathbb{1}_{\mathbb{T}})) \uparrow \eta_A$. The fact that the monad laws are given as bisimulations now implies

$$\mu(\eta(\mathbb{1}_{\mathbb{T}}) \circledcirc P) \uparrow \eta_A = P = \mu(P \circledcirc \eta(\mathbb{1}_{\mathbb{T}})) \uparrow \eta_A$$

We conclude that:

$$\begin{aligned}
\mu//A(\eta//A(\mathbb{1}_{\mathbb{T}} \uparrow \eta_A) \circledcirc P \uparrow \eta_A)\\
&= P \uparrow \eta_A\\
&= \mu//A(P \uparrow \eta_A \circledcirc \eta//A(\mathbb{1}_{\mathbb{T}} \uparrow \eta_A))
\end{aligned}$$

□

4 Polynomial Theories

4.1 Substitution Structures

Our dependent polynomial monads capture the notion of a family of dependently typed terms which are closed under substitution. However, as it stands, we have no operation allowing us to substitute terms into dependent types. The notion of substitution structure is intended to fill this gap.

Definition 4.1. Let $\mathbb{T}$ be a type structure and let M be a monad on $\mathbb{T}$, we say that M possesses a *substitution structure* if the following conditions hold:

(1) Let Γ, Δ be contexts in $\mathbb{T}$, let B be a type in $\mathbb{T}//\Delta$, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, then there exists a type $B[\sigma]$ in $\mathbb{T}//\Gamma$.

(2) Let Γ, Δ be contexts in $\mathbb{T}$, let B be a type in $\mathbb{T}//\Delta$, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, then there exists a term ρ^B_σ in $Tm_{B[\sigma],B}(M \uparrow \sigma)$.

(3) Let Λ be a context in $\mathbb{T}$, Λ' be a context in $\mathbb{T}//\Lambda$, Γ, Δ be contexts in $\mathbb{T}$, let B be a type in $\mathbb{T}//\Delta$, let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, let $\tau \in \Lambda \mapsto \Gamma$ be a substitution in M, and let b be a term in $Tm_{\Lambda',B}(M \uparrow \mu(\tau, \sigma))$, then there exists a term $b[\sigma]_\tau$ in $Tm_{\Lambda',B[\sigma]}(M \uparrow \tau)$. Furthermore, it holds that:

$$(\mu \uparrow (\tau, \sigma))(b[\sigma]_\tau, \rho^B_\sigma) = b$$

(4) Let Λ be a context in $\mathbb{T}$, Λ' be a context in $\mathbb{T}//\Lambda$, Γ, Δ be contexts in $\mathbb{T}$, let B be a type in $\mathbb{T}//\Delta$, let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, let $\tau \in \Lambda \mapsto \Gamma$ be a substitution in M, and let b be a term in $Tm_{\Lambda',B}(M \uparrow \mu(\tau, \sigma))$, then the term $b[\sigma]_\tau$ is unique in the sense that for each term b' in $Tm_{\Lambda',B[\sigma]}(M \uparrow \tau)$, fulfilling the multiplication condition in (3), it holds that $b' = b[\sigma]_\tau$.

Diagrammatically this takes the following form:

$$\begin{tikzcd}
\Lambda' \arrow[r, "b[\sigma]_\tau"] \arrow[rr, bend left, "b"] \arrow[d, Rightarrow] & B[\sigma] \arrow[r, "\rho^B_\sigma"] \arrow[d, Rightarrow] & B \arrow[d, Rightarrow] \\
\Lambda \arrow[r, "\tau"] & \Gamma \arrow[r, "\sigma"] & \Delta
\end{tikzcd}$$

Remark 4.1. As in the proof of Proposition 3.12, one can easily show that we have an induced substitution structure on $M//T$ for each type T in $\mathbb{T}$. Indeed, for two substitutions $\sigma \in \Gamma \mapsto \Delta$ and $\tau \in \Lambda \mapsto \Gamma$ in $M \uparrow \eta_T$, we simply apply the substitution structure in M to $\sigma' = \langle \sigma, \eta_T \rangle$ and $\tau' = \langle \tau, \eta_T \rangle$.

Remark 4.2. Let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, and let Δ' be a context in $\mathbb{T}//\Delta$, then we define a context $\Delta'[\sigma]$ in $\mathbb{T}//\Gamma$ by induction as follows:

- If $\Delta' = \epsilon$ then $\Delta'[\sigma] = \epsilon$.
- If $\Delta' = B \triangleright \Delta''$ for some type B in $\mathbb{T}//\Delta$ and some context Δ'' in $\mathbb{T}//\Delta//B$, then we define $\Delta'[\sigma] = B[\sigma] \triangleright \Delta''[\sigma']$ where $\sigma' = \sigma +\!\!+ \rho^B_\sigma$.

By accumulating the terms ρ^B at each step of the induction, there results a substitution

$$\xi_{\Delta'[\sigma]} : \Gamma +\!\!+ \Delta'[\sigma] \to \Gamma +\!\!+ \Delta'$$

in M.

We remark that we are aiming for a model which is strictly algebraic like that of B-systems and therefore, in addition to giving the substitutions explicity as an operation, the choice of substitution needs to be made compatible with the monad structure. In categorical models, this would correspond to the idea that "pullbacks compose strictly", not merely up to an isomorphism as guaranteed by their universal properties. We encapsulate this idea with the following definition:

Definition 4.2. Let $\mathbb{T}$ be a type structure and let M be a monad on $\mathbb{T}$, we say a substitution structure Σ is *proper* if the following conditions hold:

(1) Let Λ, Γ, Δ be contexts in $\mathbb{T}$, let $\tau \in \Lambda \mapsto \Gamma$ and $\sigma \in \Gamma \mapsto \Delta$, be substitutions in M, then

$$\mu(\rho^{C[\sigma]}_\tau, \rho^C_\sigma) = \rho^{C[\sigma]}_\tau \circledcirc \rho^C_\sigma = \rho^C_{\tau \circledcirc \sigma}$$

(2) Let Γ be a context, and let B be a type in $\mathbb{T}//\Gamma$, then

$$\rho^B_{\eta_\Gamma} = (\eta \uparrow \eta_\Gamma)(\eta_B)$$

4.2 Polynomial Theories

We now arrive at the central definition of this section.

Definition 4.3. A *polynomial theory* is a tuple $\Pi = (\mathbb{T}, M, \Sigma)$ where $\mathbb{T}$ is a type structure, M is a dependent polynomial monad on $\mathbb{T}$, and Σ is a proper substitution structure on M.

Lemma 4.4. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Then for each type T in $\mathbb{T}$ the tuple $\Pi/T = (\mathbb{T}//T, M/T, \Sigma/T)$ is a polynomial theory on $\mathbb{T}//T$.*

Proof. This is a consequence of Lemma 3.12 together with Remark 4.1. □

We obtain category of polynomial theories by defining homomorphisms as follows.

Definition 4.5. Let $\Pi = (\mathbb{T}, M, \Sigma)$ and $\Omega = (\mathbb{S}, M', \Sigma')$ be polynomial theories and let $f : \mathbb{S} \to \mathbb{T}$ be a morphism of type structures. A morphism F of the underlying polynomials relative to (f, f) is called a *homomorphism of polynomial theories* if the following conditions hold:

(1) Let A be a type in $\mathbb{S}$ then we have $F_{A,A}(\eta_A) = \eta'_{F(A)}$
(2) Let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, A be a type in $\mathbb{S}$ and $t \in M_{\Delta,A}$ then we have
$$F_{\Gamma,A}(\mu(\sigma, a)) = \mu'(F(\sigma), F(a))$$
(3) Let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, and A be a type in $\mathbb{S}//\Delta$ then we have
$$F_{A[\sigma],A}(\rho^A_\sigma) = \rho^{F(A)}_{F(\sigma)}$$

Remark 4.3. Let $f : \Pi \to \Omega$ be a homomorphism of polynomial theories. By an analogous argument as in Lemma 3.12, we can show that we can obtain a homomorphism of polynomial theories $f//T : \Pi//T \to \Omega//f_{Ty}(T)$ for each type T in $\mathbb{T}$.

Remark 4.4. A direct consequence of condition (3) of Definition 4.5 is that $F(A[\sigma]) = F(A)[F(\sigma)]$. Further since by condition (2) a homomorphism preserves multiplication, $F(a[\sigma]_\tau)$ is a mediating morphism for $F(a)$, and therefore we have, by property (4) of Definition 4.1 that $F(a[\sigma]_\tau) = F(a)[F(\sigma)]_{F(\tau)}$.

Another nice observation about this category is that we can construct morphisms of polynomial theories from substitutions in the underlying monad. This contruction will allow us to give a uniform treatment of the substitution and weakening morphisms of B-Systems in our comparison of these two structures below.

Construction 4.1. Let $\Pi = (\mathbb{S}, M, \Sigma)$ be a polynomial theory, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, we define a morphism of type structures $\hat{\sigma} : \mathbb{S}//\Delta \to \mathbb{S}$ as follows:

- For each type A in $\mathbb{S}//\Delta$ we define $\hat{\sigma}(A) = A[\sigma]$
- For each type A in $\mathbb{S}//\Delta$ we define $\hat{\sigma}/A$ as the morphism of type structures induced by the substitution $\sigma ++ \rho^A_\sigma$

Construction 4.2. Let $\Pi = (\mathbb{S}, M, \Sigma)$ be a polynomial theory, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, we define a morphism of polynomial theories $\tilde{\sigma} : \Pi//\Delta \to \Pi//\Gamma$, over $(\hat{\sigma}, \hat{\sigma})$ as follows:

- Let A be a type, and Λ be a context in $\mathbb{S}//\Delta$, further let $a \in (M//\Delta)_{\Lambda,A}$, then we define $a^* \in (M//\Gamma)_{\Lambda[\sigma],A[\sigma]}$ as
$$a^* = (\mu \uparrow (\eta_\Gamma, \sigma))(\xi_{\Lambda[A]}, a)$$
where we view $\xi_{\Lambda[A]}$ from Remark 4.2 as a substitution in $M \uparrow \sigma$. We now put $\tilde{\sigma}_{\Lambda,A}(a) = a^*[\sigma]_{\eta_\Gamma}$.
- Let A be a type, and Λ be a context in $\mathbb{S}//\Delta$, further let $a \in (M//\Delta)_{\Lambda,A}$, then we define $\tilde{\sigma} \uparrow a$ as the morphism induced by the substitution $\sigma ++ a^*$

Lemma 4.6. *The morphism of polynomial theories defined in Construction 4.2 is a homomorphism of polynomial theories.*

Proof. Conditions (1) and (2) of Definition 4.5 are direct consequences of the conditions of a proper substitution structure. Condition (3) follows from the fact that:
$$\mu(\tilde{\sigma}(\rho^A_\tau), \rho^A_\tau) = \mu(\rho^{\hat{\sigma}(A)}_{\tilde{\sigma}(\tau)}, \rho^A_\tau)$$
and therefore the terms are the same by the uniqueness condition of the substitution structure. □

5 The Equivalence of Polynomial Theories and B-Systems

We now prove that our polynomial theories are compatible with another well-known model of dependently typed theories, the *B*-systems of Voevodsky.

5.1 The Category B-Systems

We begin with a coninductive presentation inspired by [2].

Definition 5.1. A *B-Frame* $\mathbf{B}$ is defined coinductively to consist of the following data:

(1) A set $Ty_{\mathbf{B}}$ of types.
(2) For each type $A \in Ty_{\mathbf{B}}$ a set $Tm^A_{\mathbf{B}}$ of terms of type A.
(3) For each type A a new B-Frame $\mathbf{B}/A$

We think of the B-Frame $\mathbf{B}/A$ as the B-Frame of types and terms dependent on the type A.

Definition 5.2. Let $\mathbf{B}$ and $\mathbf{C}$ be B-Frames, a *morphism of B-Frames* $F : \mathbf{B} \to \mathbf{C}$ is defined coinductively as follows:

(1) A function $F_{Ty} : Ty_{\mathbf{B}} \to Ty_{\mathbf{C}}$
(2) For each type $A \in Ty_{\mathbf{B}}$ a function $F^A_{Tm} : Tm^A_{\mathbf{B}} \to Tm^{F_{Ty}(A)}_{\mathbf{C}}$
(3) For each type $A \in Ty_{\mathbf{B}}$ a morphism of B-Frames $F/A : \mathbf{B}/A \to \mathbf{C}/F_{Ty}(A)$

Remark 5.1. From now on, we will drop the subscripts and superscripts in the notation of morphisms of B-Frames, freely applying them to types and terms as appropriate.

Definition 5.3. Let $\mathbf{B}$ be a B-Frame, a *Pre-B-System* B consists of a B-Frame $\mathbf{B}$ together with:

(1) For each type $A \in Ty_{\mathbf{B}}$ a morphism $wk^{\mathsf{B}}_A : \mathbf{B} \to \mathbf{B}/A$
(2) For each type $A \in Ty_{\mathbf{B}}$ and term $a \in Tm^A_{\mathbf{B}}$ a morphism $sub_a : \mathbf{B}/A \to \mathbf{B}$
(3) For each type $A \in Ty_{\mathbf{B}}$ a generic element $\delta_A \in Tm^{wk^{\mathsf{B}}(A)}_{\mathbf{B}/A}$
(4) For each type $A \in Ty_{\mathbf{B}}$ a Pre-B-System B/A on the B-Frame $\mathbf{B}/A$

Definition 5.4. Let B and C be Pre-B-Systems, a *homomorphism* is a morphism of the underlying B-Frames $F : \mathbf{B} \to \mathbf{C}$ such that:

(1) For each type $A \in Ty_{\mathbf{B}}$ it holds that $F/A \circ wk^{\mathsf{B}}_A = wk^{\mathsf{C}}_{F(A)} \circ F$
(2) For each type $A \in Ty_{\mathbf{B}}$ and term $a \in Tm^A_{\mathbf{B}}$ it holds that $F \circ sub^{\mathsf{B}}_a = sub^{\mathsf{C}}_{F(a)} \circ (F/A)$
(3) For each type $A \in Ty_{\mathbf{B}}$ it holds that $F/A(\delta^{\mathsf{B}}_A) = \delta^{\mathsf{C}}_{F(A)}$
(4) For each type $A \in Ty_{\mathbf{B}}$ it holds that $F/A : \mathsf{B}/A \to \mathsf{C}/F_{Ty}(A)$ is a homomorphism of Pre-B-Systems.

Definition 5.5. A *B-System* $\mathfrak{B}$ is a Pre-B-System B such that the following conditions hold:

(1) For each type $A \in Ty_{\mathbf{B}}$ $wk^{\mathfrak{B}}_A$ is a homomorphism.
(2) For each type $A \in Ty_{\mathbf{B}}$ and term $a \in Tm_{\mathbf{B},A}$ $sub^{\mathfrak{B}}_a$ is a homomorphism.
(3) For each type $A \in Ty_{\mathbf{B}}$ and term $a \in Tm_{\mathbf{B},A}$ it holds that $sub^{\mathfrak{B}}_a \circ wk^{\mathfrak{B}}_A = id_{\mathfrak{B}}$.
(4) For each type $A \in Ty_{\mathbf{B}}$ $sub^{\mathfrak{B}/A}_{\delta_A} \circ wk^{\mathfrak{B}}_A/A = id_{\mathbf{B}/A}$.
(5) For each type $A \in Ty_{\mathbf{B}}$ and term $a \in Tm_{\mathbf{B},A}$ it holds that $sub_a(\delta_A) = a$.
(6) For each type $A \in Ty_{\mathbf{B}}$ the Pre-B-System $\mathfrak{B}/A$ is a B-System.

Remark 5.2. Together with these definitions, one can see that B-Systems and homomorphisms of B-Systems form a category.

Remark 5.3. Form now on we will also drop the superscripts in the notation of the Pre-B-System and B-System structure when the application is clear from the context.

Remark 5.4. Akin to type structures, there is a notion of context in B-Systems, where contexts are defined inductively as lists of types. We can then extend weakening and $B/(-)$ to contexts by recursion on them.

5.2 Polynomial Theories to B-Systems

Having set up all the definitions and notations, we can now start to build towards our main results. We start by describing how one can construct a B-System from a polynomial theory. This construction will follow the same structure as the section on B-Systems, we will start by building a B-Frame, we will then move on to constructing weakening and substitution morphisms, give a construction of generic elements and finally show that all this structure satisfies all the conditions of a B-System.

The construction of the B-Frame is quite straightforward, since we can take its types as directly generated by the type structure $\mathbb{T}$ of the polynomial theory. For the terms of the B-Frame we only consider the ones in the empty context since dependency is encoded at type level in a B-System. We finally realize the coinductive generation by slicing over the image of the identity term under η.

Construction 5.1. Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Then we define a B-Frame $\mathbf{B}$ induced by the monad M on $\mathbb{T}$ as follows:

- We let $Ty_{\mathbf{B}} = Ty_{\mathbb{T}}$
- For each type A in $\mathbb{T}$ we set $Tm^A_{\mathbf{B}} = M_{\epsilon,A}$.
- Finally for each type A in $\mathbb{T}$, we choose the B-Frame $\mathbf{B}/A$ to be the B-Frame induced by the polynomial theory $\Pi//A$ as described in Lemma 4.4.

Remark 5.5. The B-frame $\mathbf{B}$ defined here of course depends on Π, even though this is not indicated by the notation. As it will be the only B-Frame under consideration in this section, leaving out the dependence will help lighten the notation.

The next step is to define the substitution and weakening morphisms on the B-Frame $\mathbf{B}$. These will be cases of the general construction we describe in the next lemma. The main idea is to use the substitution structure Σ to build a contravariant morphism of B-Frames from a substitution $\sigma : \Gamma \mapsto \Delta$ in the polynomial M. Weakening and substitution can then be obtained from this construction as the morphisms induced by the substitutions $\bullet A$ and a respectively, where we view a term a in context ϵ as a substitution in $\epsilon \mapsto A$.

Lemma 5.6. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Further let $\mathbf{B}$ be the B-Frame induced by the monad M. Let Γ, Δ be contexts in $\mathbb{T}$, let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M then we can define a morphism of B-Frames $[\![\sigma]\!] : \mathbf{B}/\Delta \to \mathbf{B}/\Gamma$.*

Proof. We start by defining $[\![\sigma]\!]$ on types. Let C be a type in $Ty_{\mathbf{B}/\Delta}$, which is by definition a type in $\mathbb{T}/\Delta$. We define $[\![\sigma]\!](C) \coloneqq C[\sigma]$, as given in condition (1) of the substitution structure Σ, which is a type in $\mathbb{T}/\Gamma$ and therefore also in $Ty_{\mathbf{B}/\Gamma}$.

Now let c be a term in $Tm^C_{\mathbf{B}/\Delta}$, so $c \in (M \uparrow \eta_\Delta)_{\epsilon,C}$, we start by defining

$$c^* \coloneqq (\mu \uparrow (\sigma, \eta_\Delta))(\bullet\epsilon, c).$$

By choosing $\Lambda = \Gamma$, $\Lambda' = \epsilon$ and $\tau = \eta_\Gamma$ in condition 3 of the substitution structure Σ we obtain a term $c^*[\sigma]_{\eta_\Gamma}$ in $Tm^{\mathbf{B}/\Gamma}_{C[\sigma]}$, since $c^*[\sigma]_{\eta_\Gamma} \in (P \uparrow \eta_\Gamma)_{\epsilon,C[\sigma]}$.

Finally, we define $[\![\sigma]\!]/D$ as $[\![\sigma +\!\!+ \rho^D_\sigma]\!]$, repeating the construction above. □

Showing that the construction of Lemma 5.6 respects composition will allow us to reduce the proof that the B-System conditions hold for our constructed B-Frame to cases of the properness of our substitution structure, since it allows us to reason about the compositions of substitution and weakening as compositions of substitutions in the polynomial theory. This compatibility is recorded in the following Lemma, whose proof we leave to the Appendix.

Lemma 5.7. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory and let $\mathbf{B}$ be the B-Frame induced by the monad M. Further Let Λ, Γ, Δ be contexts in $\mathbb{T}$, let $\tau \in \Lambda \mapsto \Gamma$ and $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, then it holds that:*

$$[\![\tau]\!] \circ [\![\sigma]\!] = [\![\tau \circledcirc \sigma]\!]$$

Remark 5.6. The variance here may look suprising at first glance given the contravariance of $[\![-]\!]$. The difference is due to the fact that we write composition of subsitutions in a polynomial in diagrammatic order so that this contravariance is hidden.

Lemma 5.8. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Further let $\mathbf{B}$ be the B-Frame induced by the monad M, now let A be a type in $Ty_{\mathbf{B}}$ and let a be a term in $Tm^A_{\mathbf{B}}$, then we can define morphisms of B-Frames $wk_A : \mathbf{B} \to \mathbf{B}/A$ and $sub_a : \mathbf{B}/A \to \mathbf{B}$.*

Proof. This is a direct application of Lemma 5.6, choosing $\Gamma = A$, $\Delta = \epsilon$ and $\sigma = \bullet A \in A \mapsto \epsilon$ for wk_A, and $\Gamma = \epsilon$, $\Delta = A$ and $\sigma = a \in \epsilon \mapsto A$ for sub_a. □

The generic element is build from the unit of the monad. Intuitively this is what we would expect since composition with the unit returns the original term and substituting into the variable or substituting the variable into a term returns the term itself. For the substitution structure to be applicable to η_A we need to view η_A as a term in $(M \uparrow \bullet A)_{\epsilon,A}$, which is possible by Construction 3.3.

Lemma 5.9. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Further let $\mathbf{B}$ be the B-Frame induced by the monad M, then we can define a generic element on $\mathbf{B}$, where wk is defined as in Lemma 5.8.*

Proof. Let A be a type in $Ty_{\mathbf{B}}$, by Construction 3.3 we know that η_A corresponds to a term η'_A in $(M \uparrow \bullet A)_{\epsilon,A}$. Then we define $\delta_A \coloneqq \eta'_A[\bullet A]_{\eta_A}$, which is a term in $Tm^{\mathbf{B}/A}_{\epsilon,A[\bullet A]}$. Diagrammatically this takes the following form:

$$\begin{tikzcd}
\epsilon \arrow[r, "\delta_A"] \arrow[d, Rightarrow] \arrow[rr, bend left, "\eta'_A"] & A[\bullet A] \arrow[r, "\rho^A_{\bullet A}"] \arrow[d, Rightarrow] & A \arrow[d, Rightarrow] \\
A \arrow[r, "\eta_A"] & A \arrow[r, "\bullet\epsilon"] & \epsilon
\end{tikzcd}$$

□

Obtaining the Pre-B-System is now just a matter of combining all the previous results.

Proposition 5.10. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Then we can define a Pre-B-System $\mathtt{B}$ on the B-Frame induced by M.*

The final step is to show that all the B-System conditions hold for our constructed Pre-B-System. As we already mentioned before this will be a consequence of the substitution structure Σ being proper. The main argument behind this reduction is that by Lemma 5.7 we know that $[\![-]\!]$ respects composition of substitutions, we will use this in the proof that $[\![-]\!]$ is a homomorphism below. The proof of the Proposition is long, but relatively straightforward so we include it in the Appendix.

Proposition 5.11. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory and let $\mathbf{B}$ be the B-Frame induced by the monad M. Further Let Γ, Δ be contexts in $\mathbb{T}$, let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M, then $[\![\sigma]\!]$ is a homomorphism of B-Systems.*

Using the proposition, we can now show:

Theorem 5.12. *Let $\Pi = (\mathbb{T}, M, \Sigma)$ be a polynomial theory. Then the Pre-B-System $\mathfrak{B}$ defined on the B-Frame induced by M is a B-System.*

Proof. Since wk and sub are both defined using $[\![-]\!]$ as in Lemma 5.6, it follows from Proposition 5.11 that both wk and sub are homomorphisms of B-Systems. This shows that Properties (1) and (2) hold. Properties (3) and (4) of Definition 5.5 follow by the same argument as in the proof of Proposition 5.11, together with condition (2) in the definition of properness since the involved substitutions compose to the identity substitution. Finally, condition (5) of Definition 5.5 is a consequence of the fact that in this case, $(\delta_A)^* = a$ and so we again obtain the desired result from condition (2) of Definition 4.2. □

Constructing homorphisms of B-Systems from homorphisms of polynomial theories is more straightforward, since we can extract the morphsims by the same construction we used in Section 5.2, that is, only considering terms in context ϵ and slicing only along the terms η_A for each type A. The homomorphism properties then are a straightforward application of the homomorphism properties of polynomial theories.

Lemma 5.13. *Let $(F, f) : \Pi \to \Omega$ be a morphism of polynomial theories, then there is an induced morphism of B-Systems*

$$\hat{F} : \mathfrak{B}_\Pi \to \mathfrak{B}_\Omega$$

Proof. We define $\hat{F}_{Ty} = f_{Ty}$. For each type A in $Ty_{\mathfrak{B}_\Pi}$ we define $\hat{F}_{A,Tm} = F_{\epsilon,A}$. Finally, let A be a type in $Ty_{\mathfrak{B}_\Pi}$, then we define $\hat{F}/A$ as the morphism of B-Systems induced by the homomorphism of polynomial theories $F \uparrow \eta_A$. □

Lemma 5.14. *The morphism defined in Lemma 5.13 is a homomorphism of B-Systems.*

Proof. Property (1) and (2) of Definition 5.4 are a direct consequence of Remark 4.4. Property (3) is a consequence of the same Remark, combined with the fact that F preserves η by condition (2) of Definition 4.5. Finally, property (4) is a consequence of Remark 4.3. □

Remark 5.7. Combining Lemma 5.13 with Theorem 5.12 we obtain a functor from the category of polynomial theories to the category of B-Systems. Which we will denote as $\mathscr{F} : \mathtt{PolyTh} \to \mathtt{Bsys}$.

5.3 B-Systems to Polynomial Theories

5.3.1 B-Systems to Dependent Polynomials. The reverse construction, that is, producting a polynomial theory from a B-system, is somewhat more involved than the previous one. Consequently, many proofs in the section will be given in the Appendix.

Lemma 5.15. *A B-System $\mathfrak{B}$ has an underlying type structure $\mathbb{T}$.*

Proof. We choose $Ty_{\mathbb{T}} = Ty_{\mathbf{B}}$. Further, let A be a type in $Ty_{\mathbb{T}}$, then we choose the type structure $\mathbb{T}//A$ to be the type structure induced by the B-System $\mathfrak{B}/A$. □

It turns out to be convenient to associate a dependent polynomial to a *homomorphism* f of B-systems, rather that a single B-system $\mathfrak{B}$. This is because the slicing operation of a polynomial can return a polynomial over two different type structures, even when starting from a polynomial over a single type structure. Consequently, the coinductive step must be generalized. The dependent polynomial associated to a B-system $\mathfrak{B}$ can then be obtained by applying this construction to the identity homomorphism of $\mathfrak{B}$.

Lemma 5.16. *Let $\mathfrak{B}$ and $\mathfrak{C}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ be a homomorphism of B-Systems. Then the triple $(\mathfrak{B}, \mathfrak{C}, f)$ induces a dependent polynomial $P : \mathbb{T}_{\mathfrak{B}} \to \mathbb{T}_{\mathfrak{C}}$ where $\mathbb{T}_{\mathfrak{B}}$ and $\mathbb{T}_{\mathfrak{C}}$ are the type structures induced by $\mathfrak{B}$ and $\mathfrak{C}$ respectively.*

Proof. Let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$ and let A be a type in $\mathbb{T}_{\mathfrak{C}}$, we define $P_{\Gamma,A}$ as the set $Tm^{\mathfrak{B}/\Gamma}_{(wk_\Gamma \circ f)(A)}$. Now let a be a term in $P_{\Gamma,A}$. We start by defining a homomorphism of B-Systems $\Phi_a : \mathfrak{C}/A \to \mathfrak{B}/\Gamma$ by setting

$$\Phi_a \coloneqq sub_a \circ (wk_\Gamma \circ f)/A$$

We then take $P \uparrow a$ as the dependent polynomial induced by the triple $(\mathfrak{B}/\Gamma, \mathfrak{C}/A, \Phi_a)$. The fact that Φ_a follows from the assumption that f is. □

Remark 5.8. We will use Φ_a to denote the homomorphism of B-Systems $\Phi_a : \mathfrak{C}/A \to \mathfrak{B}/\Gamma$ defined in the proof of Lemma 5.16 from now on.

5.3.2 B-Systems to Polynomial Monads. We will start this section by constructing the monad transformations η and μ on the dependent polynomial P induced by a B-System $\mathfrak{B}$ by an application of Lemma 5.16 to the identity homomorphism of $\mathfrak{B}$. Afterward we will show that these transformations respect the monad identities, and we can therefore assemble the presented data into a monad.

Lemma 5.17. *Let $\mathfrak{B}$ be a B-System, then $\mathfrak{B}$ and let $P : \mathbb{T} \to \mathbb{T}$ be the dependent polynomial induced by $\mathfrak{B}$, then there is a polynomial morphism $\eta : \mathbb{1}_{\mathbb{T}} \to P$.*

Before we are able to prove Lemma 5.17 we need the following intermediary lemma.

Lemma 5.18. *Let $\mathfrak{B}$ be a B-System, let $P : \mathbb{T} \to \mathbb{T}$ be the dependent polynomial induced by $\mathfrak{B}$ and let A be a type in $\mathbb{T}$, therefore a type in $Ty_{\mathfrak{B}}$, then $P \uparrow \delta_A = P_{\mathfrak{B}/A}$.*

Proof. We know from Lemma 5.16 that $P \uparrow \delta_A$ is defined as the polynomial induced by the triple $(\mathfrak{B}/A, \mathfrak{B}/A, \Phi_{\delta_A})$, where $\Phi_{\delta_A} : \mathfrak{B}/A \to \mathfrak{B}/A$ is defined as $sub_{wk_A(\delta_A)} \circ (wk_A/A)$. Now by property 4 of the B-System $\mathfrak{B}$ we have that $\Phi_{\delta_A} = id_{\mathfrak{B}/A}$, therefore the dependent polynomial is induced by the triple $(\mathfrak{B}/A, \mathfrak{B}/A, id_{\mathfrak{B}/A})$ which is by definition $P_{\mathfrak{B}/A}$. □

Using Lemma 5.18 we can now prove Lemma 5.17

Proof. Let Γ be a context in $\mathbb{T}$ and let A be a type in $\mathbb{T}$, we need to define a function

$$\eta_{\Gamma,A} : (\mathbb{1}_{\mathbb{T}})_{\Gamma,A} \to P_{\Gamma,A}.$$

Where we can assume without loss of generality that $\Gamma = A$, then by definition of $\mathbb{1}_{\mathbb{T}}$ we have that $\mathbb{1}_{\mathbb{T}_{A,A}} = \{\mathbb{1}_A\}$. Therefore, we only need to define η_A, which we choose to be δ_A as given by the generic element on $\mathfrak{B}$, so $\eta_A = \delta_A$.

Now let a be a term in $\mathbb{1}_{\mathbb{T}_{A,A}}$. We need to define a morphism of dependent polynomials

$$\eta \uparrow a : \mathbb{1}_{\mathbb{T}} \uparrow a \to P \uparrow \eta_{\Gamma,A}(a).$$

But by Lemma 5.18 we have that $P \uparrow \eta_A = P_{\mathfrak{B}/A}$, and we can again assume, without loss of generality that $a = \mathbb{1}_A$. Further, by definition of $\mathbb{1}_{\mathbb{T}}$ we have that $\mathbb{1}_{\mathbb{T}} \uparrow a = \mathbb{1}_{\mathbb{T}/A}$, so we can repeat the previous construction on the B-System $\mathfrak{B}/A$ to obtain the desired morphism of dependent polynomials. □

The case of the multiplication μ will be an result from a chain multiple lemmas. The idea behind the construction is that if we have a substitution $\sigma \in \Gamma \mapsto \Delta$ in the induced polynomial, we can build a morphism of B-Systems Θ_σ from $\mathfrak{B}/\Delta$ to $\mathfrak{B}/\Gamma$ using σ. This substitution will be build from repeated applications of the construction Φ_a of Remark 5.8. We then show that this Θ_σ is always a homomorphism of B-Systems which in turn allows us to define $\mu(\sigma, t) = \Theta_\sigma(t)$ and show that the resulting term is indeed typed correctly.

This strategy, as well as a verification of the monad laws, is carried out in the Appendix. In particular, the construction of Θ_σ and its properties appears as Lemma 6.2.

Lemma 5.19. *Let $\mathfrak{B}$ be a B-System, let $P : \mathbb{T} \nrightarrow \mathbb{T}$ be the dependent polynomial induced by $\mathfrak{B}$, then there is a polynomial morphism $\mu : P \circledcirc P \to P$*

Lemma 5.20. *Let $\mathfrak{B}$ be a B-System, and $\mathbb{T}$ be the type structure induced by $\mathfrak{B}$, then $\mathfrak{B}$ induces a polynomial monad M on $\mathbb{T}$.*

Proof. Let $P : \mathbb{T} \nrightarrow \mathbb{T}$ be the dependent polynomial induced by $\mathfrak{B}$ and let $\eta : \mathbb{1}_{\mathbb{T}} \to P$ and $\mu : P \circledcirc P \to P$ be the polynomial morphisms as given in Lemma 5.17 and Lemma 5.19, then by Lemma 6.6 and Remark 6.4 we have that the tuple (P, η, μ) forms a polynomial monad on $\mathbb{T}$. □

5.3.3 B-Systems to Polynomial Theories. Finally, we construct a substitution structure on the monad we obtained from a B-System, this structure will allow us to account for substituting terms into types. It will on types be given as application of the previously defined Θ-morphisms, while on terms its action will be trivial. The action on terms being trivial is a consequence of the fact that in the polynomial induced by a B-System, the terms a in context Γ, of type A, over some substitution σ are exactly the terms in context Γ of type $\Theta_\sigma(A)$. The term ρ can then be chosen to be the general element of $A[\sigma]$.

Lemma 5.21. *Let $\mathfrak{B}$ be a B-System. Then we can define a substitution structure on the monad as given in Lemma 5.20.*

Proof. Let Γ, Δ be contexts in $\mathbb{T}$, let B be a type in $\mathbb{T}//\Delta$, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in M we define $B[\sigma] \coloneqq \Theta_\sigma(B)$, we further define $\rho^B_\sigma \coloneqq \delta_{\Theta_\sigma(B)}$. Now let Λ be a context in $\mathbb{T}$, Λ' be a context in $\mathbb{T}//\Lambda$, let $\tau \in \Lambda \mapsto \Gamma$ be a substitution in M, and let b be a term in $(M \uparrow \mu(\tau, \sigma))_{\Lambda',B}$, we define $b[\sigma]_\tau \coloneqq b$, which is correct, since the type of b is the following:

$$wk_{\Lambda'} \circ \Theta_{\tau\circ\sigma}(B) = wk_{\Lambda'} \circ \Theta_\tau(\Theta_\sigma(B)) = wk_{\Lambda'} \circ \Theta_\tau(B[\sigma])$$

Now we consider:

$$\begin{aligned}
(\mu \uparrow (\tau, \sigma))(b[\sigma]_\tau, \rho^B_\sigma) &= \Theta_{b[\sigma]_\tau}(\rho^B_\sigma) \\
&= \Theta_{b[\sigma]_\tau}(\delta_{\Theta_\sigma(B)}) \\
&= sub^{\mathfrak{B}/\Lambda ++ \Lambda'}_{b[\sigma]_\tau} \circ (wk^{\mathfrak{B}/\Lambda}_{\Lambda'} \circ \Theta_\tau)/B[\sigma](\delta_{\Theta_\sigma}) \\
&= sub^{\mathfrak{B}/\Lambda ++ \Lambda'}_{b[\sigma]_\tau}(\delta_{wk^{\mathfrak{B}/\Lambda}_{\Lambda'} \circ \Theta_\tau \circ \Theta_\sigma(B)}) \\
&= b[\sigma]_\tau = b
\end{aligned}$$

The fact that $b[\sigma]_\tau$ is unique, is a direct consequence of property (5) of Definition 5.5, since we have for two terms b_1, b_2 of type B with $sub_{b_1}(\delta_B) = sub_{b_2}(\delta_B)$ that $b_1 = b_2$ holds.
Finally, to obtain a substitution structure on $M//T$ we just repeat the construction above on $\mathfrak{B}/T$. □

That the substitution structure defined above is proper will be a consequence of the properties of a B-System. That the substitution structure respects composition will follow from the fact that the Θ-morphisms respect composition, together with the fact that homomorphisms of B-Systems map generic elements to generic elements. The fact that substitution along the identity is the identity will be a consequence $\Phi_{\delta_A} = Id_{\mathfrak{B}/A}$ and therefore $\Theta_{\eta_\Gamma} = Id_{\mathfrak{B}/\Gamma}$.

Lemma 5.22. *Let $\mathfrak{B}$ be a B-System, then the substitution structure defined in Lemma 5.21 is proper.*

Proof. Let Λ, Γ, Δ be contexts in $\mathbb{T}$, let $\sigma \in \Lambda \mapsto \Gamma$, $\tau \in \Gamma \mapsto \Delta$, be substitutions in M, then we have:

$$\begin{aligned}
&(\mu \uparrow (\sigma, \tau))(\rho_\sigma^{C[\tau]}, \rho_\tau^C)\\
=&\Theta_{\rho_\sigma^{C[\tau]}}(\rho_\tau^C)\\
=&\Theta_{\delta_{\Theta_\sigma(C[\tau])}}(\delta_{\Theta_\tau(C)})\\
=&sub^{\mathfrak{B}/\Lambda ++ C[\tau][\sigma]}_{\delta_{\Theta_\sigma(C[\tau])}} \circ (wk^{\mathfrak{B}/\Lambda}_{C[\tau][\sigma]} \circ \Theta_\sigma)/C[\tau](\delta_{\Theta_\tau(C)})\\
=&sub^{\mathfrak{B}/\Lambda ++ C[\tau][\sigma]}_{\delta_{\Theta_\sigma(C[\tau])}}(\delta_{wk^{\mathfrak{B}/\Lambda}_{C[\tau][\sigma]} \circ \Theta_\sigma \circ \Theta_\tau(C)})\\
=&\delta_{\Theta_\sigma(C[\tau])} = \delta_{\Theta_\sigma(\Theta_\tau(C))} = \delta_{\Theta_{\sigma\circ\tau}(C)} = \rho^C_{\sigma \circledcirc \tau}
\end{aligned}$$

The second property follows directly from the fact that $\Theta_{\eta_\Gamma} = Id_{\mathfrak{B}/\Gamma}$. □

Finally, since a B-System induces a monad with a proper substitution structure on it, we can conclude that it induces a polynomial theory.

Theorem 5.23. *A B-System $\mathfrak{B}$ induces a polynomial theory $\Pi_{\mathfrak{B}}$.*

5.3.4 Morphisms of induced Polynomial Theories.

Lemma 5.24. *Let $f : \mathfrak{B} \to \mathfrak{C}$ be a homomorphism of B-Systems, then there is an induced morphism of type structures $\tilde{f} : \mathbb{T}_{\mathfrak{B}} \to \mathbb{T}_{\mathfrak{C}}$.*

Proof. We take $\tilde{f}_{Ty} = f_{Ty}$ and for each $A \in Ty_{\mathfrak{B}}$ we define $\tilde{f}_A$ as the morphism induced by $f//A$. □

As we did in Section 5.3 we will need to generalize our construction of the functorial action on morphisms to the case of a commuting square. This is again an artifact of the coinductive presentation: as we slice a polynomial, the type structures start to diverge from each other. In order to complete the coinductive definition on morphisms, therefore, we need additional data, recovering the action on a single morphism as a special case. A correct statement is the following:

Lemma 5.25. *Let $\mathfrak{B}, \mathfrak{C}, \mathfrak{D}, \mathfrak{E}$ be B-Systems, and let f, g, h, k be homomorphisms of B-Systems such that the following diagram commutes:*

$$\begin{tikzcd}
\mathfrak{B} \arrow[d, "h"] & \mathfrak{C} \arrow[l, "f"'] \arrow[d, "k"] \\
\mathfrak{D} & \mathfrak{E} \arrow[l, "g"]
\end{tikzcd}$$

Then we can define a morphism of polynomials F, from the polynomial induced by the triple $(\mathfrak{B}, \mathfrak{C}, f)$ to the polynomial induced by the triple $(\mathfrak{D}, \mathfrak{E}, g)$ over the morphisms of type structures induced by h and k.

Proof. Let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$ and let A be a type in $\mathbb{T}_{\mathfrak{C}}$, we define $F_{\Gamma,A} = h/\Gamma \circ wk^{\mathfrak{B}}_\Gamma \circ f$. Now let a be a term in $P_{\Gamma,A}$, we define $F \uparrow a$ as the morphism of polynomials induced by the homomorphism of B-Systems $f = sub_a \circ (wk^{\mathfrak{B}}_\Gamma \circ f)/A$, $g = sub_{F_{\Gamma,A}(a)} \circ (wk^{\mathfrak{D}}_{h(\Gamma)} \circ g)/k(A)$, $k = k/A$ and $h = h/\Gamma$. □

Lemma 5.26. *Let $v : \mathfrak{B} \to \mathfrak{C}$ be a homomorphism of B-Systems, then there is an induced morphism of the polynomials induced by $\mathfrak{B}$ and $\mathfrak{C}$ over the type structures defined in Lemma 5.24.*

Proof. This is a special case of Lemma 5.25 by taking $\mathfrak{B} = \mathfrak{C}$, $\mathfrak{D} = \mathfrak{E}$, $g = id_{\mathfrak{D}}$, $h = v = k$ and $f = id_{\mathfrak{B}}$. □

Remark 5.9. Let $f : \mathfrak{B} \to \mathfrak{C}$ be a homomorphism of B-Systems, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in $\Pi_{\mathfrak{B}}$, then we define $f(\sigma) \in f(\Gamma) \mapsto f(\Delta)$ by recursion on σ as follows:

- If $\sigma = \bullet\Gamma$ then $f(\sigma) = \bullet f(\Gamma)$
- if $\sigma = \langle t, \sigma' \rangle$ then for some term $t \in (P_{\mathfrak{B}})_{\Gamma',B}$ and substitution $\sigma' \in \Gamma' \mapsto \Delta'$ then we define $f(\sigma) = \langle f/\Gamma'(t), f/\Gamma'(\sigma') \rangle$

Lemma 5.27. *Let $f : \mathfrak{B} \to \mathfrak{C}$ be a homomorphism of B-Systems, and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in $\Pi_{\mathfrak{B}}$, then it holds that*

$$f/\Gamma \circ \Theta_\sigma = \Theta_{f(\sigma)} \circ f/\Delta$$

Proof. Again, we show this by induction over σ: Assume that $\sigma = \bullet\Gamma$, then we have: $f/\Gamma \circ \Theta_\sigma = f/\Gamma \circ wk^{\mathfrak{B}}_\Gamma = wk^{\mathfrak{C}}_{f(\Gamma)} \circ f = \Theta_{f(\sigma)} \circ f/\Delta$. Since f is a homomorphism of B-Systems.
Otherwise, we know that $\Theta_\sigma = wk_{\Gamma'} \circ sub_a \circ (wk_{\Gamma''} \circ \Theta_{\sigma'})/A$, for some substitution $\sigma' \in \Gamma''' \mapsto \Delta'$, type A in $\mathfrak{B}//\Delta'$ and term $a \in (P_{\mathfrak{B}})_{\Gamma'',A}$. Where $\Gamma = \Gamma''' ++ \Gamma'' ++ \Gamma'$ and $\Delta = \Delta' ++ A$. Now be repeated application of the homomorphism properties of f, together with the induction hypothesis we have:

$$\begin{aligned}
&f/\Gamma \circ \Theta_\sigma\\
&= f/\Gamma \circ (wk_{\Gamma'} \circ sub_a \circ (wk_{\Gamma''} \circ \Theta_{\sigma'})/A)\\
&= wk_{f(\Gamma'')(\Gamma')} \circ f/\Gamma'' \circ sub_a \circ (wk_{\Gamma''} \circ \Theta_{\sigma'})/A\\
&= wk_{f(\Gamma'')(\Gamma')} \circ sub_{f/\Gamma''(a)} \circ (f/\Gamma'' \circ wk_{\Gamma''} \circ \Theta_{\sigma'})/A\\
&= wk_{f(\Gamma'')(\Gamma')} \circ sub_{f/\Gamma''(a)} \circ (wk_{f/\Gamma'''(\Gamma'')} \circ f/\Gamma''' \circ \Theta_{\sigma'})/A\\
&= wk_{f(\Gamma'')(\Gamma')} \circ sub_{f/\Gamma''(a)} \circ (wk_{f/\Gamma'''(\Gamma'')} \circ \Theta_{f(\sigma')} \circ f/\Delta')/A\\
&= \Theta_{f(\sigma)} \circ f/\Delta
\end{aligned}$$

□

Lemma 5.28. *The morphism defined in Lemma 5.26 is a homomorphism of polynomial theories.*

Proof. Since f is a homomorphism it holds that:

$$F_{A,A}(\eta_A) = f/A(\delta_A) = \delta_{f(A)} = \eta'_{F(A)}$$

That F preserves μ follows directly from Lemma 5.27. Finally, that F preserves ρ is a direct consequence of the previous two conditions, since $\rho^A_\sigma = \delta_{\Theta_\sigma(A)}$. □

Remark 5.10. In light of the previous Lemma, we now have a functor $\mathcal{G} : \texttt{BSys} \to \texttt{PolyTh}$.

5.4 B-Systems and Polynomial Theories are equivalent

It remains to show that the categories are equivalent, we aim to show this by showing that the functor $\mathcal{F}$ is essentially surjective and fully faithful. This will be done by showing that for any B-System $\mathfrak{B}$ we have that $\mathcal{F}(\mathcal{G}(\mathfrak{B})) = \mathfrak{B}$ and that for any homomorphism of B-Systems $f : \mathfrak{B} \to \mathfrak{C}$ it holds that $\mathcal{F}(\mathcal{G}(f)) = f$. Since $\mathcal{F}(\mathcal{G}(\mathfrak{B})) = \mathfrak{B}$ implies essential surjectivity and $\mathcal{F}(\mathcal{G}(f)) = f$ implies fullness, the proof of the equivalence then reduces to showing that $\mathcal{F}$ is faithful. This will be a consequence of the fact that all terms $a \in (P \uparrow \sigma)_{\Gamma,A}$ in a polynomial theory are in one to one correspondence to terms in $((P \uparrow \sigma) \uparrow \bullet\Gamma)_{\epsilon,A[\bullet\Gamma]}$ and therefore

the restriction of the polynomial morphism to terms in context ϵ already contains enough information to distinguish morphisms of polynomial theories.

Lemma 5.29. *Let $\mathfrak{B}$ be a B-System, then it holds that $\mathcal{F}(\mathcal{G}(\mathfrak{B})) = \mathfrak{B}$.*

Proof. We start by showing that the underlying B-Frames of the two B-Systems are equal: Let $Ty_{\mathbf{B}}$ be the set of types in $\mathbf{B}$ and let $Ty_{\mathbb{T}}$ be the set of types in $\mathbb{T}$, then by definition of $\mathcal{G}$ and $\mathcal{F}$ we have that $Ty_{\mathcal{F}(\mathcal{G}(\mathbf{B}))} = Ty_{\mathbb{T}} = Ty_{\mathbf{B}}$.

The same holds for the sets of terms in a type $Tm^A_{\mathfrak{B}}$ since the terms in $Tm^A_{\mathcal{F}(\mathcal{G}(\mathbf{B}))}$ are defined as the ones in $\mathcal{G}(\mathfrak{B})_{\epsilon,A}$ which are exactly the terms in $Tm^A_{\mathfrak{B}}$ by definition of $\mathcal{G}$.

Now consider the B-Frame $\mathbf{B}/A$ for a type $A \in Ty_{\mathbf{B}}$, by definition of $\mathcal{G}$ and $\mathcal{F}$ we have that the B-Frame $\mathcal{F}(\mathcal{G}(\mathbf{B}))/A$ is defined as the B-Frame induced by the dependent polynomial $P \uparrow \eta_A = P \uparrow \delta_A$. This is the polynomial induced by the triple $(\mathfrak{B}/A, \mathfrak{B}/A, \Phi_{\delta_A})$ as in Lemma 5.16, but $\Phi_{\delta_A} = sub^{\mathfrak{B}/A}_{(\delta_A)} \circ (wk^{\mathfrak{B}}_A/A) = id_{\mathbf{B}/A}$ by property 4 of Definition 5.5 and therefore the B-Frame induced by $P \uparrow \delta_A$ is equal to $\mathbf{B}/A$ itself.

It remains to show that the additional structure of the B-Systems are equal. For the morphisms, we will restrict ourselves to showing that the substitution morphisms are equal, the proof for the weakening morphisms is similar.

For this, let Δ be an arbitrary context in $\mathbb{T}//A$,and let C be a type in $\mathbb{T}//\Delta$. Then we know that $sub^{\mathcal{F}(\mathcal{G}(\mathfrak{B}))}_a/\Delta$ is defined as $[\![\xi_{\Delta[a]}]\!]$, then we know that:

$$c^* = (\mu \uparrow \tilde{\mu}(\eta_{\Delta[a]}, \xi_{\Delta[a]}))(\epsilon, \bullet\epsilon, c)$$

so, $c^* = \Theta_{\xi_{\Delta[a]}}(c)$, but unfolding the coinductive definition of $\xi_{\Delta[a]}$ and using property (4) of Definition 5.5, we know that $\Theta_{\xi_{\Delta[a]}} = sub^{\mathfrak{B}}_a/\Delta$, therefore by Lemma 5.21 we have that $c[a] = c^* = sub^{\mathfrak{B}}_a/\Delta(c)$. The same holds for types, since $B[\sigma] = \Theta_\sigma(B)$ for any substitution σ.

The fact that $\delta^{\mathcal{F}(\mathcal{G}(\mathfrak{B}))}_A = \delta^{\mathfrak{B}}_A$ is immediate from the definitions, since we chose $\delta^{\mathcal{F}(\mathcal{G}(\mathfrak{B}))}_A = \eta_A[\bullet A]_{\eta_A} = \delta^{\mathfrak{B}}_A$. □

Lemma 5.30. *Let $f : \mathfrak{B} \to \mathfrak{C}$ be a homomorphism of B-Systems, then it holds that $\mathcal{F}(\mathcal{G}(f)) = f$.*

Proof. In the base case we have $\mathcal{F}(\mathcal{G}(f))_{Ty} = f_{Ty}$ and

$$\mathcal{F}(\mathcal{G}(f))_{A,Tm} = id \circ wk_\epsilon \circ f_{A,Tm} = f_{A,Tm}.$$

For the inductive step we have that $\mathcal{F}(\mathcal{G}(f))/A$ is the homomorphism of B-Systems induced by $h = id_{\mathfrak{B}}/f(A)$, $k = id_{\mathfrak{C}}/A$, $f = sub_{\delta_f(A)} \circ (wk_{f(A)} \circ f)/A = f/A$ and $g = sub_{\delta_{g(A)}} \circ (wk_{g(A)} \circ g)/A = g/A$. So we can repeat the argument. □

Lemma 5.31. *The functor $\mathcal{F}$: `PolyTh` $\to$ `BSys` is fully faithful.*

Proof. Fullness is a consequence of Lemma 5.30. To show that $\mathcal{F}$ is faithful, let $f, g : \Pi \to \Omega$ be morphisms of polynomial such that $\mathcal{F}(f) = \mathcal{F}(g)$, further assume that $f \neq g$. Let $\mathbb{T}$ and $\mathbb{S}$ be the type structures underlying Π and Ω respectively. If $f \neq g$ then there exist contexts Γ in $\mathbb{T}$ and Δ in $\mathbb{S}$, context Λ in $\mathbb{S}//\Gamma$, a type A in $\mathbb{S}//\Delta$, a substitution $\sigma : \Gamma \to \Delta$ and a term $a \in (M \uparrow \sigma)_{\Lambda,A}$ such that $f \uparrow \sigma(a) \neq g \uparrow \sigma(a)$, and $\mathcal{F}(f)(\mathcal{F}(a)) = \mathcal{F}(g)(\mathcal{F}(a))$. But $\mathcal{F}(a) = a[\bullet\Lambda]_{\eta_\Lambda}$ and therefore $f \uparrow (\sigma +\!\!+ \bullet\Lambda)(a[\bullet\Lambda]_{\eta_\Lambda}) \neq g \uparrow (\sigma +\!\!+ \bullet\Lambda)(a[\bullet\Lambda]_{\eta_\Lambda})$, which is a contradiction since f and g preserve the substitution structure. □

The preceding lemmas now show

Theorem 5.32. *The functor $\mathcal{F}$: `PolyTh` $\to$ `BSys` forms an equivalence of categories.*

References

[1] Jiří Adámek and Jiří Rosickỳ. 2001. On sifted colimits and generalized varieties. *Theory and Applications of Categories [electronic only]* 8 (2001), 33–53.

[2] Benedikt Ahrens, Jacopo Emmenegger, Paige Randall North, and Egbert Rijke. 2025. Algebraic Presentations of Type Dependency. *Logical Methods in Computer Science* 21 (2025).

[3] Steve Awodey. 2018. Natural models of homotopy type theory. *Mathematical Structures in Computer Science* 28, 2 (2018), 241–286.

[4] Aurelio Carboni and Peter Johnstone. 1995. Connected limits, familial representability and Artin glueing. *Mathematical Structures in Computer Science* 5, 4 (1995), 441–459.

[5] John Cartmell. 1986. Generalised algebraic theories and contextual categories. *Annals of pure and applied logic* 32 (1986), 209–243.

[6] G. Cruttwell and M. Shulman. 2010. A Unified Framework for Generalized Multicategories. *Theory and Applications of Categories* 24, 21 (2010), 580–655.

[7] Peter Dybjer. 1995. Internal type theory. In *International Workshop on Types for Proofs and Programs*. Springer, 120–134.

[8] Nicola Gambino, Richard Garner, and Christina Vasilakopoulou. 2025. A unified treatment of commuting tensor products of categories, operads, symmetric multicategories and their bimodules. *arXiv preprint arXiv:2511.14402* (2025).

[9] Nicola Gambino and André Joyal. 2017. *On operads, bimodules and analytic functors*. Vol. 249. American Mathematical Society.

[10] Richard Garner. 2015. Combinatorial structure of type dependency. *Journal of Pure and Applied Algebra* 219, 6 (2015), 1885–1914. doi:10.1016/j.jpaa.2014.07.015

[11] Bart Jacobs. 1993. Comprehension categories and the semantics of type dependency. *Theoretical Computer Science* 107, 2 (1993), 169–207.

[12] Joachim Lambek. 2006. Deductive systems and categories II. Standard constructions and closed categories. In *Category Theory, Homology Theory and their Applications I: Proceedings of the Conference held at the Seattle Research Center of the Battelle Memorial Institute, June 24–July 19, 1968 Volume One*. Springer, 76–122.

[13] Tom Leinster. 2004. *Higher operads, higher categories*. Number 298. Cambridge University Press.

[14] Egbert Rijke et al. 2025. The Agda-Unimath Library. https://unimath.github.io/agda-unimath/ See specifically the subchapter on dependent type theories..

[15] Michael Shulman. 2008. Framed bicategories and monoidal fibrations. *Theory and Applications of Categories [electronic only]* 20 (2008), 650–738.

[16] Vladimir Voevodsky. 2014. B-systems. *arXiv preprint arXiv:1410.5389* (2014).

6 Appendix

6.1 Polynomial Theories to B-Systems

Proof of Lemma 5.11. Since both wk and sub are defined using $[\![-]\!]$ and we know by Lemma 5.7 that $[\![-]\!]$ respects composition of substitutions, conditions (1) and (2) of Definition 5.4 reduce to checking that the compositions of the involved substitutions are equal. We show the first condition, the second is similar. Let A be a type in $Ty_{\mathbf{B}/\Delta}$, then we need to show that:

$$[\![\sigma]\!]/A \circ wk_A = wk_{[\![\sigma]\!](A)} \circ [\![\sigma]\!]$$

Unfolding the definitions, we get:

$$\begin{aligned}&[\![\sigma +\!\!+ \rho^A_\sigma]\!] \circ [\![\eta_\Delta +\!\!+ \bullet A]\!]\\ &= [\![\eta_\Gamma +\!\!+ \bullet A[\sigma]]\!] \circ [\![\sigma]\!]\end{aligned}$$

Which reduces to:

$$[\![(\sigma +\!\!+ \rho^A_\sigma) \circledcirc (\eta_\Delta +\!\!+ \bullet A)]\!] = [\![(\eta_\Gamma +\!\!+ \bullet A[\sigma]) \circledcirc \sigma]\!]$$

Now since

$$\sigma \circledcirc \eta_\Delta = \sigma = \eta_\Gamma \circledcirc \sigma$$

and

$$\rho^A_\sigma \circledcirc \bullet A = \bullet A[\sigma]$$

we obtain that both sides are equal to $\sigma +\!\!+ \rho^A_\sigma$ and therefore the equality holds.

Now for condition (3) of Definition 5.4, let A be a type in $Ty_{\mathbf{B}/\Delta}$, then we need to show that:

$$[\![\sigma]\!]/A(\delta_A) = \delta_{[\![\sigma]\!](A)}$$
$$[\![\sigma +\!\!+ \rho^A_\sigma]\!](\delta_A) = \delta_{A[\sigma]}$$

First, we note that by definition of $\delta_{A[\sigma]}$ in Lemma 5.9, we have that, for any term in $a[\sigma]_\tau$ in $(M \uparrow \eta_\Gamma)_{\Lambda, A[\sigma]}$ it holds that:

$$\begin{aligned}
&\mu(\mu(\langle \bullet\epsilon, a[\sigma]_\tau\rangle, (\eta_{A[\sigma]} +\!\!+ \delta_{A[\sigma]})))\\
&, (\rho^{A[\sigma]}_{\bullet A[\sigma]} +\!\!+ \bullet A[\sigma])\\
&= \mu(a[\sigma]_\tau, \eta_{A[\sigma]})\\
&= a[\sigma]_\tau
\end{aligned}$$

Where we wrote μ as a shorthand for $\mu \uparrow (\eta_\Gamma, \eta_\Gamma)$ to improve readability.

From here on we will argue this point diagrammatically where horizontal arrows represent substitutions and vertical arrows represent type dependency. Concatenating arrows represents application of μ and in squares, the upper arrow represents a substitution over a substitution represented by the lower arrow. Now, we know that $\delta_{A[\sigma]}$ is unique with this property, since $\eta_{A[\sigma]}$ is unique. Further, we note that

$$(\delta_A)^* = (\mu \uparrow (\sigma, \eta_\Gamma))(\rho^A_\sigma, \eta_A) = \rho^A_\sigma$$

and therefore that the following diagram composes to a^*:

$$\begin{tikzcd}
\epsilon \arrow[r] \arrow[d, Rightarrow] & \epsilon \arrow[r, "(\delta_A)^*[\sigma +\!\!+ \rho^A_\sigma]"] \arrow[d, Rightarrow] \arrow[rr, bend left, "(\delta_A)^*"] & A[\bullet A][\sigma] \arrow[r, "\rho^{A[\bullet A]}_{\sigma +\!\!+ \rho^A_\sigma}"] \arrow[d, Rightarrow] & A[\bullet A] \arrow[r, "\rho^A_{\bullet A}"] \arrow[d, Rightarrow] & A \arrow[d, Rightarrow] \\
\Lambda' \arrow[r, "a^*[\sigma]^*"] \arrow[d, Rightarrow] & A[\sigma] \arrow[r] \arrow[d, Rightarrow] & A[\sigma] \arrow[r, "\rho^A_\sigma"] \arrow[d, Rightarrow] & A \arrow[r, "\bullet\epsilon"] \arrow[d, Rightarrow] & \epsilon \arrow[d, Rightarrow] \\
\Lambda \arrow[r, "\tau"] & \Gamma \arrow[r] & \Gamma \arrow[r, "\sigma"] & \Delta \arrow[r] & \Delta
\end{tikzcd}$$

But then, by condition (3) of Definition 4.1, the following diagram also composes to a^{**}:

$$\begin{tikzcd}
\epsilon \arrow[r] \arrow[d, Rightarrow] & \epsilon \arrow[r, "(\delta_A)^*[\sigma +\!\!+ \rho^A_\sigma]"] \arrow[d, Rightarrow] & A[\sigma][\bullet A[\sigma]] \arrow[r, "\rho^{A[\bullet A]}_{\eta_\Gamma +\!\!+ \bullet A[\sigma]}"] \arrow[d, Rightarrow] & A[\sigma] \arrow[r, "\rho^A_\sigma"] \arrow[d, Rightarrow] & A \arrow[d, Rightarrow] \\
\Lambda' \arrow[r, "a^*[\sigma]^*"] \arrow[d, Rightarrow] & A[\sigma] \arrow[r] \arrow[d, Rightarrow] & A[\sigma] \arrow[r, "\bullet A[\sigma]"] \arrow[d, Rightarrow] & \epsilon \arrow[r, "\bullet\epsilon"] \arrow[d, Rightarrow] & \epsilon \arrow[d, Rightarrow] \\
\Lambda \arrow[r, "\tau"] & \Gamma \arrow[r] & \Gamma \arrow[r] & \Gamma \arrow[r, "\sigma"] & \Delta
\end{tikzcd}$$

Since we know that $[\![\sigma]\!]$ commutes with wk. But then, since the mediating terms in the substitution structure are unique, we obtain that the following diagram composes to $a^*[\sigma]^*$:

$$\begin{tikzcd}
\epsilon \arrow[r] \arrow[d, Rightarrow] & \epsilon \arrow[r, "(\delta_A)^*[\sigma +\!\!+ \rho^A_\sigma]"] \arrow[d, Rightarrow] & A[\sigma][\bullet A[\sigma]] \arrow[r, "\rho^{A[\bullet A]}_{\eta_\Gamma +\!\!+ \bullet A[\sigma]}"] \arrow[d, Rightarrow] & A[\sigma] \arrow[r, "\rho^A_\sigma"] \arrow[d, Rightarrow] & A \arrow[d, Rightarrow] \\
\Lambda' \arrow[r, "a^*[\sigma]^*"] \arrow[d, Rightarrow] & A[\sigma] \arrow[r] \arrow[d, Rightarrow] & A[\sigma] \arrow[r, "\bullet A[\sigma]"] \arrow[d, Rightarrow] & \epsilon \arrow[r, "\bullet\epsilon"] \arrow[d, Rightarrow] & \epsilon \arrow[d, Rightarrow] \\
\Lambda \arrow[r, "\tau"] & \Gamma \arrow[r] & \Gamma \arrow[r] & \Gamma \arrow[r, "\sigma"] & \Delta
\end{tikzcd}$$

And therefore by the uniqueness of $\delta_{A[\sigma]}$ we obtain that

$$[\![\sigma +\!\!+ \rho^A_\sigma]\!](\delta_A) = \delta_{A[\sigma]}.$$

Lastly, condition (4) of Definition 5.4 follows by repeating this construction on the appended substitution. □

Proof of Lemma 5.7. Let A be a type in $\mathbb{T}//\Delta$, since M is proper, we know that

$$A[\sigma][\tau] = A[\sigma \circledcirc \tau]$$

Now let a be a term in $Tm^A_{\mathbf{B}//\Delta}$, so $a \in (M \uparrow \eta_\Delta)_{\epsilon, A}$, then we need to show that:

$$(a^*[\sigma]_{\eta_\Gamma})^*[\tau]_{\eta_\Lambda} = a^{**}[\sigma \circledcirc \tau]_{\eta_\Lambda}$$

where we write a^{**} to distinguish the term we obtain during the application of $[\![\sigma \circledcirc \tau]\!]$. Now since μ is associative, we know that

$$\begin{aligned}
&(\mu \uparrow (\tau, \sigma))((a^*[\sigma]_{\eta_\Gamma})^*, \rho^A_\sigma)\\
=&(\mu \uparrow (\tau, \sigma))((\mu \uparrow (\tau, \eta_\Gamma))(\bullet\epsilon, (a^*[\sigma]_{\eta_\Gamma})), \rho^A_\sigma)\\
=&(\mu \uparrow (\tau, \sigma))(\bullet\epsilon, (\mu \uparrow (\eta_\Gamma, \sigma))(a^*[\sigma]_{\eta_\Gamma}, \rho^A_\sigma))\\
=&a^{**}
\end{aligned}$$

And therefore, since now both $(a^*[\sigma]_{\eta_\Gamma})^*[\tau]_{\eta_\Lambda}$ and $a^{**}[\sigma \circledcirc \tau]_{\eta_\Lambda}$ are mediating in the sense of condition (3) of the substitution structure Σ, we obtain that $(a^*[\sigma]_{\eta_\Gamma})^*[\tau]_{\eta_\Lambda} = a^{**}[\sigma \circledcirc \tau]_{\eta_\Lambda}$. □

6.2 B-Systems to Polynomial Theories

6.2.1 The construction of μ. Before we are able to prove Lemma 5.19 we need to be able to transform substitutions in P into homomorphisms of B-Systems and show some algebraic properties of these homomorphisms. We do this in the next lemmas. The proof of Lemma 5.19 will then reduce to a special case of Lemma 6.5.

Lemma 6.1. *Let $\mathfrak{B}$ and $\mathfrak{C}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ be a homomorphism of B-Systems. Then $\Phi_a \circ wk^{\mathfrak{C}}_A = wk^{\mathfrak{B}}_\Gamma \circ f$ for each term a in $Tm^{(wk_\Gamma \circ f)(A)}_{\mathfrak{B}/\Gamma}$ with A being a type in $\mathbb{T}_{\mathfrak{C}}$.*

Proof. We have that $\Phi_a = sub_a \circ (wk_\Gamma \circ f)/A$, and therefore:

$$\begin{aligned}
\Phi_a \circ wk^{\mathfrak{C}}_A &= sub_a \circ (wk^{\mathfrak{B}}_\Gamma \circ f)/A \circ wk^{\mathfrak{C}}_A\\
&= sub_a \circ wk^{\mathfrak{B}/\Gamma}_{(wk^{\mathfrak{B}}_\Gamma \circ f)Ty(A)} \circ wk^{\mathfrak{B}}_\Gamma \circ f\\
&= wk^{\mathfrak{B}}_\Gamma \circ f
\end{aligned}$$

where the second equality follows from the fact that f is a homomorphism of B-Systems, and therefore $wk_\Gamma \circ f$ is a homomorphism of B-Systems. The last equality follows from property (3) of B-Systems. □

Remark 6.1. In the special case where $\mathfrak{B} = \mathfrak{C}$ and $f = id_{\mathfrak{B}}$, Lemma 6.1 reduces to the statement that $\Phi_a \circ wk^{\mathfrak{B}}_A = wk^{\mathfrak{B}}_\Gamma$ for each term a in $Tm^{wk_\Gamma(A)}_{\mathfrak{B}/\Gamma}$.

The construction below is the core of the multiplication. As we already described above the idea is to use the fact that since a substitution is a list of terms, which depend on the previous terms in the list, we can use the morphism built by the repeated application of Φ_a along this list to obtain a homomorphism of B-Systems.
This results in a proof which looks unintuitive at first sight since the construction of the homomorphism is hidden in the definition of the dependent polynomial, which is why we take the time to explain the idea again here.

Lemma 6.2. *Let $\mathfrak{B}$ and $\mathfrak{C}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ be a homomorphism of B-Systems. Further let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$ and Δ be a context in $\mathbb{T}_{\mathfrak{C}}$ and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in the dependent polynomial induced by the triple $(\mathfrak{B}, \mathfrak{C}, f)$, then there exists a homomorphism of B-Systems. $\Theta_\sigma : \mathfrak{C}/\Delta \to \mathfrak{B}/\Gamma$*

Proof. We do this by induction over the structure of σ. If $\sigma = \bullet\Gamma$ then we choose $\Theta_\sigma = wk^{\mathfrak{B}}_{\Gamma} \circ f$. Now let $\sigma = \langle t, \sigma' \rangle$ for a term t in $P_{\Gamma',B}$ and a substitution $\sigma' : \Gamma'' \mapsto \Delta'$ in $P \uparrow t$, where $\Gamma = \Gamma' +\!\!+ \Gamma''$ and $\Delta = B \triangleright \Delta'$. By the induction hypothesis we have a homomorphism of B-Systems $\Theta_{\sigma'} : (\mathbf{C}/B)/\Delta' \to (\mathbf{B}/\Gamma')/\Gamma''$, which is a homomorphism $\Theta_{\sigma'} : \mathbf{C}/\Delta \to \mathbf{B}/\Gamma$. We can now define Θ_σ as

$$\Theta_\sigma = \Theta_{\sigma'}.$$

□

The following Lemmas are algebraic identities about the composition of Θ_σ with combinations of morphisms as they will appear in the construction of μ. We need the identities to show that the term we construct is typed correctly, since terms in the dependent polynomial induced by triples of the form $(\mathfrak{B}, \mathfrak{C}, f)$ are obtained by applying f, we need to show that an application of Θ_σ reduces to the morphism which induces the composed polynomial.

Lemma 6.3. *Let $\mathfrak{B}$ and $\mathfrak{C}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ be a homomorphism of B-Systems. Further let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$ and Δ be a context in $\mathbb{T}_{\mathfrak{C}}$ and let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in the dependent polynomial induced by the triple $(\mathfrak{B}, \mathfrak{C}, f)$, then $\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} = wk^{\mathfrak{B}}_{\Gamma} \circ f$*

Proof. We show this by induction on Δ. If $\Delta = \epsilon$ then by definition $\Theta_\sigma = wk^{\mathfrak{B}}_{\Gamma} \circ f$, so the statement holds.
Now let $\Delta = B \triangleright \Delta'$ for a type B in $\mathbb{T}_{\mathfrak{C}}$ and a context Δ' in $\mathbb{T}_{\mathfrak{C}}//B$, then by definition $\sigma = \langle t, \sigma' \rangle$ for a term t in $P_{\Gamma',B}$ and a substitution $\sigma' : \Gamma'' \mapsto \Delta'$ in $P \uparrow t$, where $\Gamma = \Gamma' +\!\!+ \Gamma''$. By the induction hypothesis we have that $\Theta_{\sigma'} \circ wk^{\mathfrak{C}}_{\Delta'} = wk^{\mathfrak{B}}_{\Gamma''} \circ \Phi_t$. Then we have that

$$\begin{aligned}
\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} &= \Theta_{\sigma'} \circ wk^{\mathfrak{C}}_{\Delta'} \circ wk^{\mathfrak{C}}_{B} \\
&= wk^{\mathfrak{B}/\Gamma'}_{\Gamma''} \circ \Phi_b \circ wk^{\mathfrak{C}}_{B} \\
&= wk^{\mathfrak{B}/\Gamma'}_{\Gamma''} \circ wk^{\mathfrak{B}}_{\Gamma'} \circ f \\
&= wk^{\mathfrak{B}}_{\Gamma} \circ f
\end{aligned}$$

Where we used Lemma 6.1 in the third equality. □

Remark 6.2. In the special case where $\mathfrak{B} = \mathfrak{C}$ and $f = id_{\mathbf{B}}$ this reduces to $\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} = wk^{\mathfrak{B}}_{\Gamma}$. Since the polynomial induced by $\mathfrak{B}$ is defined to be the one induced by the triple $(\mathfrak{B}, \mathfrak{B}, id_{\mathbf{B}})$.

Lemma 6.4. *Let $\mathfrak{B}$, $\mathfrak{C}$ and $\mathfrak{D}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ and $g : \mathfrak{D} \to \mathfrak{C}$ be homomorphisms of B-Systems. Further let $P : \mathbb{T}_{\mathfrak{B}} \to \mathbb{T}_{\mathfrak{C}}$ and $Q : \mathbb{T}_{\mathfrak{C}} \to \mathbb{T}_{\mathfrak{D}}$ be the dependent polynomials induced by the triples $(\mathfrak{B}, \mathfrak{C}, f)$ and $(\mathfrak{C}, \mathfrak{D}, g)$ respectively. Let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$, Δ be a context in $\mathbb{T}_{\mathfrak{C}}$, $\sigma \in \Gamma \mapsto \Delta$ be a substitution in P and t be a term in $Q_{\Delta,A}$ for a type A in $\mathbb{T}_{\mathfrak{D}}$, then it holds that*

$$\Theta_\sigma \circ \Phi_a = \Phi_{\Theta_\sigma(a)}$$

Proof.

$$\begin{aligned}
\Theta_\sigma \circ \Phi_a &= \Theta_\sigma \circ sub^{\mathfrak{C}}_{wk^{\mathfrak{C}}_{\Delta}(a)} \circ (wk^{\mathfrak{C}}_{\Delta} \circ g)/A \\
&= sub^{\mathfrak{B}}_{wk^{\mathfrak{B}}_{\Gamma}(\Theta_\sigma(a))} \circ \Theta_\sigma/((wk^{\mathfrak{C}}_{\Delta} \circ g)(A)) \\
&\circ (wk^{\mathfrak{C}}_{\Delta} \circ g)/A \\
&= sub^{\mathfrak{B}}_{wk^{\mathfrak{B}}_{\Gamma}(\Theta_\sigma(a))} \circ (\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} \circ g)/A \\
&= \Phi_{\Theta_\sigma(a)}
\end{aligned}$$

Where we used the fact that Θ_σ is a homomorphism of B-Systems, since f is one. □

Like in the case of defining the dependent polynomial from a B-System, we will first define the general case of Lemma 5.19 for dependent polynomials induced by triples of B-Systems and homomorphisms between them. We will then obtain Lemma 5.19 as a special case.
The construction itself is an application of the theory developed in the previous lemmas, we will realize multiplication as application of Θ_σ and then use the algebraic identities shown above to prove that the resulting term is typed correctly and that when we construct $\Phi_{\Theta_\sigma(t)}$, we obtain the correct morphism to induce the sliced polynomial.

Lemma 6.5. *Let $\mathfrak{B}$, $\mathfrak{C}$ and $\mathfrak{D}$ be B-Systems and let $f : \mathfrak{C} \to \mathfrak{B}$ and $g : \mathfrak{D} \to \mathfrak{C}$ be homomorphisms of B-Systems. Further let $P : \mathbb{T}_{\mathfrak{B}} \nrightarrow \mathbb{T}_{\mathfrak{C}}$ and $Q : \mathbb{T}_{\mathfrak{C}} \nrightarrow \mathbb{T}_{\mathfrak{D}}$ be the dependent polynomials induced by the triples $(\mathfrak{B}, \mathfrak{C}, f)$ and $(\mathfrak{C}, \mathfrak{D}, g)$ respectively. Then we can define a polynomial morphism*

$$\mu : P \circledcirc Q \to R$$

where $R : \mathbb{T}_{\mathfrak{B}} \nrightarrow \mathbb{T}_{\mathfrak{D}}$ is the dependent polynomial induced by the triple $(\mathfrak{B}, \mathfrak{D}, f \circ g)$.

Proof. Let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$ and let A be a type in $\mathbb{T}_{\mathfrak{D}}$, we need to define a function

$$\mu_{\Gamma,A} : (P \circledcirc Q)_{\Gamma,A} \to R_{\Gamma,A}.$$

We note that by definition of composition of dependent polynomials, a term in $(P \circledcirc Q)_{\Gamma,A}$ is of the form (Δ, σ, t) where Δ is a context in $\mathbb{T}_{\mathfrak{C}}$, $\sigma \in \Gamma \mapsto \Delta$ is a substitution in P and t is a term in $Q_{\Delta,A}$. We define $\mu(\sigma, t) = \Theta_\sigma(t)$, since we know by Lemma 6.3 that $\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} = wk^{\mathfrak{B}}_{\Gamma} \circ f$ we know that the type of $\Theta_\sigma(t)$ is:

$$\Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} \circ g(A) = (wk^{\mathfrak{B}}_{\Gamma} \circ f) \circ g(A)$$

and therefore $\Theta_\sigma(t)$ is a term in $R_{\Gamma,A}$.
Further let (Δ, σ, t) be a term in $(Q \circ P)_{\Gamma,A}$, we need to define a morphism of dependent polynomials

$$\mu \uparrow (\Delta, \sigma, t) : (Q \circ P) \uparrow (\Delta, \sigma, t) \to R \uparrow \mu_{\Gamma,A}(\Delta, \sigma, t).$$

But since $(Q \circ P) \uparrow (\Delta, \sigma, t) = Q \uparrow \sigma \circ P \uparrow t$ and $R \uparrow \mu_{\Gamma,A}(\Delta, \sigma, t) = R \uparrow \Theta_\sigma \circ wk^{\mathfrak{B}}_{\Delta} \circ g(t)$ we claim that this can be achieved by repeating the previous construction on the tuple

$$(\mathfrak{B}/\Gamma, \mathfrak{C}/\Delta, \mathfrak{D}/A, \Theta_\sigma, \Phi_t).$$

This holds true if R^* induced by the triple $(\mathfrak{B}/\Gamma, \mathfrak{D}/A, \Theta_\sigma \circ \Phi_t)$ is equal to $R \uparrow \mu_{\Gamma,A}(\Delta, \sigma, t)$.
By Lemma 5.16 we know that $R \uparrow \mu_{\Gamma,A}(\Delta, \sigma, t)$ is induced by the triple $(\mathfrak{B}/\Gamma, \mathfrak{D}/A, \Phi_{\mu_{\Gamma,A}(\Delta,\sigma,t)})$, so it suffices to show that $\Phi_{\mu_{\Gamma,A}(\Delta,\sigma,t)} = \Theta_\sigma \circ \Phi_t$. But since $\Phi_{\mu_{\Gamma,A}(\Delta,\sigma,t)} = \Phi_{\Theta_\sigma(t)}$ this is an application of Lemma 6.4. □

Remark 6.3. Now we obtain Lemma 5.19 from Lemma 6.5 by choosing $\mathfrak{B} = \mathfrak{C} = \mathfrak{D}$ and $f = g = id_{\mathbf{B}}$.

6.3 The Monad Laws

The unit law for the monad structure follows the intuition we described when we explained the choice of δ_A as η_A. The unit law is now a consequence of the interactions between the substitution morphism and the generic element δ_A.

Lemma 6.6. *Let $\mathfrak{B}$ be a B-System, let $P : \mathbb{T} \nrightarrow \mathbb{T}$ be the dependent polynomial induced by $\mathfrak{B}$, then for $\eta : \mathbb{1}_{\mathbb{T}} \to P$ and $\mu : P \circ P \to P$ as given in Lemma 5.17 and 5.19 it holds that $\mu(P \circ \eta(\mathbb{1}_{\mathbb{T}})) = P = \mu(\eta(\mathbb{1}_{\mathbb{T}}) \circ P)$.*

Proof. We start with the base case, let A be a type in $\mathbb{T}$, Γ be a context in $\mathbb{T}$ and let a be a term in $P_{\Gamma,A}$.
Then in the first case, we have

$$\begin{aligned}\mu(a, \eta_A) &= \Phi_a(\eta_A)\\ &= sub_a \circ wk_\Gamma/A(\delta_A)\\ &= sub_a(\delta_{wk_\Gamma(A)})\\ &= a\end{aligned}$$

Where we used property (3) of Definition5.4 in the third equality and property (5) of Definition 5.5 in the last equality.
In the second case, we get:

$$\begin{aligned}\mu(\eta_\Gamma, a) &= \Theta_{\eta_\Gamma}(a)\\ &= id_{\mathfrak{B}/\Gamma}(a)\\ &= a\end{aligned}$$

By the argument in Lemma 5.18 it suffices to consider this base case, since we can conclude that this is a bisimulation by repeating the argument on the polynomial induced by the B-System $\mathfrak{B}/A$ for each type A in $Ty(\mathfrak{B})$ while we have to replace $wk^{\mathfrak{C}}_{\Gamma}/A$ with $(wk^{\mathfrak{C}}_{\Gamma} \circ g)/A$ in the first argument.

□

Before we are able to show that μ is associative, we need to be able to show that if we compose two morphisms, induced by substitutions, the resulting morphism is the one induced by the composed substitution. So we show that the construction of Θ_σ is compatible with composition of substitutions.

Lemma 6.7. *Let $P = (\mathfrak{B}, \mathfrak{C}, f)$ and $Q = (\mathfrak{C}, \mathfrak{D}, g)$ be dependent polynomials induced by B-Systems and let $R = (\mathfrak{B}, \mathfrak{D}, f \circ g)$ be the dependent polynomial induced by composing the homomorphisms of B-Systems, further let $\mu : Q \circ P \to R$ be the polynomial morphism as given in Lemma 6.5. Now let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$, Δ be a context in $\mathbb{T}_{\mathfrak{C}}$, Ω be a context in $\mathbb{T}_{\mathfrak{D}}$, let $\sigma \in \Gamma \mapsto \Delta$ and $\tau \in \Delta \mapsto \Omega$ be substitutions in P and Q respectively, then it holds that $\Theta_{\sigma \circledcirc \tau} = \Theta_\sigma \circ \Theta_\tau$.*

Proof. We do this by recursion on τ, as in Lemma 3.11. If $\tau = \bullet\Delta$ then $\Theta_\tau = wk^{\mathfrak{C}}_{\Delta} \circ g$ and therefore

$$\begin{aligned}\Theta_{\tau \circledcirc \sigma} &= \Theta_{\bullet\Gamma}\\ &= wk^{\mathfrak{B}}_{\Gamma} \circ f \circ g\\ &= \Theta_\sigma \circ wk^{\mathfrak{C}}_{\Delta} \circ g\\ &= \Theta_\sigma \circ \Theta_\tau\end{aligned}$$

Now let $\tau = \langle t, \tau' \rangle$ for some term t in $Q_{\Delta',T}$ where T is a type in $\mathbb{T}_{\mathfrak{D}}$ and $\tau' : \Delta'' \mapsto \Omega'$ is a substitution with $\Delta = \Delta' +\!\!+ \Delta''$ and $\Omega = T \triangleright \Omega'$. Further we get substitutions $\sigma' : \Gamma' \mapsto \Delta'$ and $\sigma'' : \Gamma'' \mapsto \Delta''$ with $\Gamma = \Gamma' +\!\!+ \Gamma''$ and $\sigma = \sigma' +\!\!+ \sigma''$. By the induction hypothesis we have that $\Theta_{\tau' \circ \sigma''} = \Theta_{\sigma''} \circ \Theta_{\tau'}$, and therefore we get:

$$\begin{aligned}\Theta_{\sigma \circledcirc \tau} &= \Theta_{\langle \mu(\sigma', t), \sigma'' \circledcirc \tau' \rangle}\\ &= \Theta_{\sigma'' \circledcirc \tau'}\\ &= \Theta_{\sigma''} \circ \Theta_{\tau'}\\ &= \Theta_\sigma \circ \Theta_\tau\end{aligned}$$

Since by definition if $\Theta_\tau = \Theta_{\langle t, \tau' \rangle}$ then $\Theta_\tau = \Theta_{\tau'}$, which generalizes to the case of $\sigma = \sigma' +\!\!+ \sigma''$. □

Since we defined μ by applying the homomorphisms of B-Systems induced by substitutions, the associativity of μ follows from the fact that Θ_σ respects the composition of substitutions. Further, since we can extend μ naturally to substitutions, as shown in Lemma 3.11, we give the proof of associativity for triples (σ, τ, t) instead of working with substitutions in $P \circledcirc Q$ directly, since if we have a substitution in $P \circledcirc Q$, we can build σ by vertical composition of substitutions, as given in Construction 3.4.

Lemma 6.8. *Let $\mathfrak{B}, \mathfrak{C}, \mathfrak{D}$ and $\mathfrak{E}$ be B-Systems, let $f : \mathfrak{C} \to \mathfrak{B}$, $g : \mathfrak{D} \to \mathfrak{C}$ and $h : \mathfrak{E} \to \mathfrak{D}$ be homomorphisms of B-Systems. Further let $P : \mathbb{T}_{\mathfrak{B}} \to \mathbb{T}_{\mathfrak{C}}$, $Q : \mathbb{T}_{\mathfrak{C}} \to \mathbb{T}_{\mathfrak{D}}$ and $R : \mathbb{T}_{\mathfrak{D}} \to \mathbb{T}_{\mathfrak{E}}$ be the dependent polynomials induced by the triples $(\mathfrak{B}, \mathfrak{C}, f)$, $(\mathfrak{C}, \mathfrak{D}, g)$ and $(\mathfrak{D}, \mathfrak{E}, h)$ respectively. Finally, let μ be defined as in Lemma 6.5, then it holds that*

$$\mu(\mu(P \circledcirc Q) \circledcirc R) = \mu(P \circledcirc \mu(Q \circledcirc R))$$

Proof. Let Γ be a context in $\mathbb{T}_{\mathfrak{B}}$, Δ be a context in $\mathbb{T}_{\mathfrak{C}}$, Ω be a context in $\mathbb{T}_{\mathfrak{D}}$, let $\sigma \in \Gamma \mapsto \Delta$ be a substitution in P, let $\tau \in \Delta \mapsto \Omega$ be a substitution in Q and let t be a term in $R_{\Omega,A}$ for a type A in $\mathbb{T}_{\mathfrak{E}}$. Then we get:

$$\begin{aligned}\mu(\sigma, \mu(\tau, t)) &= \Theta_\sigma \circ \Theta_\tau(t)\\ &= \Theta_{\sigma \circ \tau}(t)\\ &= \mu(\mu(\sigma, \tau), t)\end{aligned}$$

Where we used Lemma 6.7 in the third equality. □

Remark 6.4. The fact that μ as defined in Lemma 5.19 is associative is now a special case of the previous lemma by choosing $\mathfrak{B} = \mathfrak{C} = \mathfrak{D} = \mathfrak{E}$ and $f = g = h = id_{\mathbf{B}}$.